

The Just-in-Time World

The Roundtable · Study 3 · the complete study, print edition · 84 sections, 235 register rows

The metacrisis at the level of stuff: modern civilization stands on six raw materials — sand, salt, iron, copper, oil, lithium — moved through supply chains tuned for efficiency over resilience. Which links in those chains actually break; when one does, how long before the shelves, the grid, and the fab lines feel it; and which disruptions pass in weeks versus which redraw the map for good — reported as honest bands, not one scare number. (After Ed Conway's Material World.)

How to read a tag

A boxed tag such as `row-id · validated` names the register row a sentence rests on and that row's standing. A dotted tag is a pointer or a ruling, not a claim's support. Appendix A lists every row; Appendix B lists every engine, its seed, and who rebuilt it.

How we know any of this

HOW WE KNOW ANY OF THIS

Four times in this study, at four different links, the same institutional thing happens. An aquifer under the Atacama is drawn down for two decades before a court orders the baseline study that would have measured it. A refining complex dating from the early twentieth century begins reporting what crosses its fence line in 2019. The chemical plants that make the plastic in an IV bag were required to begin monitoring for vinyl chloride at the property line in July 2026, and on the day the requirement took effect, a proclamation exempted a named list of them, including some of the largest, for two more years `chl-g1 · validated`. Disposal wells that had been raising the ground pressure under West Texas for years are required, from 2023, to carry a gauge deep in the hole and report what it reads. The intervals run from about five years to more than a century, and in every case the instrument that would let the people living there see what is happening to them is built *after* the harm is established, not before `study-g1 · validated`.

That much is only what you would expect, and on its own it is a complaint rather than a finding. The part worth noticing is what kind of instrument gets built. All four are public. The benzene readings from every American refinery are on an EPA dashboard; the Texas injection pressures go to a university-run tool open to anyone; the Chilean settlement is supervised in open court with annual compliance hearings, and the chemical-plant rule is the refinery rule's descendant by EPA's own account. **None of these had to be.** A regulator can measure privately and act on what it finds, and the harm would be addressed just the same. Building the instrument in public does something else: it hands the people who live there a standing capability they did not have before, and one that does not depend on the regulator's continued attention. **Each quarter of refinery fence-line data carries a date by which it must be public, and a quarter once published cannot be unpublished.** An enforcement action ends; a data stream keeps producing. What that changes is who is able to make the argument, not who wins it. Port Arthur's fence line has been publishing **since 2019**, and Port Arthur's air is still Port Arthur's air.

The boundary of the pattern is visible inside this study too. Around the Guixi smelter in Jiangxi, the arsenic and cadmium in the soil and the vegetables were established by university researchers and an NGO, not by an instrument anyone in those villages can consult. So the claim is narrower than it first looks, and worth stating exactly: **where a regulator eventually acts, the thing it builds is public, and where none does, the knowledge exists without belonging to anyone who could use it.**

Copper — the wire in the wall

This is an outlet. There are about a dozen within reach of where you're sitting, and nothing about it suggests it has a history.

It's screwed into a wall; say, in a new house outside Dallas. The wire behind its faceplate runs into the stud bay, up through the plate, and back through every room to a gray breaker panel in the garage.

And behind that panel, the wire keeps going: back through a lumberyard's distributor, a Georgia rod mill, two container ports and an ocean, a Chinese smelter, a port with its own desalination plant, and a hole in the driest desert on Earth. Half a world of arrangements, all so the light comes on.

This study is about those arrangements, and about the fact that almost none of them keep any slack.

CHAPTER ONE · COPPER

Follow one spool of building wire, start to finish. Its paperwork would look something like this; the study fills in each line only as the referees validate it.

WHAT IT FEELS LIKE

Your builder calls: the framing is up, the house is on hold, the wire has not come. You ask how long. Nobody can tell you: what a distributor keeps is in no survey this study could find, and it looked. Nobody's lights go out. The wire in your walls crossed the Pacific twice before it got there.

dfw-b1 · validated; col-d1 · validated; rh-bunker-comp · validated

1 · EXTRACT

The wire begins as ore under the Atacama Desert at **Minera Escondida**: one pit complex producing about one tonne in every eighteen of the world's mined copper (FY2025) esc-c2 · validated. Its concentrate leaves not by truck but by a 170-kilometer slurry pipeline to the mine's own private port col-f1 · validated.

Across Chile, concentrating copper took about 13 gigajoules of electricity for every tonne of copper in the concentrate in 2024, up by nearly half since 2010. The state copper commission gives the reason plainly: deeper pits and harder rock. In the open pits, the fuel burned per tonne of copper has doubled over the same years as ore grades fell esc-f5 · validated.

This is the district at true scale: pits, leach ponds, and tailings works spanning tens of kilometers. The water that keeps it running is seawater now, desalinated at the coast and pumped 3,100 meters uphill esc-c1 · validated. The basin's Indigenous communities (the Peine community filed before the state did) took the aquifer drawdown to court; the case ended in a court-approved conciliation (fourteen measures, a standing governance table, implementation still under tribunal supervision), while the court put the aquifer's own recovery at around the year 2200 esc-g1 · validated.

Lay a familiar outline over the district, at the same scale: Compare: Manhattan

2 · PROCESS

The concentrate ships across the Pacific to **Guixi**, in Jiangxi province, the only single plant on Earth that refines more than a million tonnes a year [gux-c2 · validated](#). The wire's copper becomes 99.99% cathode here.

The smelter at true scale, rail spurs running into it, and farming villages in its deposition plume, about fifteen villages, some ten thousand people, with documented metals in their soil and vegetables [gux-f1 · validated](#). **What powers the smelter is not on public record.** A 1,880-megawatt, four-unit coal station stands beside it (Global Energy Monitor's figures); whether any of that output crosses the fence is stated nowhere a reader can check [rh-gux-power-p · validated](#). The dossier card carries who, energy, ecosystem, and one comparison.

3 · COMBINE

Then the copper crosses the same ocean again: cathode east to Long Beach, then overland to Carrollton, Georgia, where **Southwire** can cast it to rod at close to a tonne a minute [swr-f3 · validated](#) and its Building Wire Plant in the same town makes the wire [swr-f5 · validated](#). The insulation extruded around it is PVC: the salt chain, entering the copper chain [salt-x2 · validated](#).

THE CROSSING

day oof 14 to 15, this distance at the container fleet's own average speed [route-t1 · validated](#)

Keep going. The ship is doing the same thing you are: nothing, for a fortnight, at about the speed of a bicycle.

Every other beat of this study compresses a leg into a sentence. This one does not, because the compression is the lie. **Waiting is what this chain mostly does.** This crossing is the shorter of the two: the ore's run to the smelter takes **more than twice as long** [route-t3 · validated](#), and the two together are roughly 48 to 52 days at sea, with nothing to report and nobody able to hurry it.

The fuel is the only thing being spent: four or five gallons of bunker for this spool's share of the sea legs [rh-bunker-comp · validated](#), which is the oil chapter, running underneath this one.

When a chain like this breaks, this is the part that cannot be made up: you can restart a plant, and you cannot shorten an ocean.

One Southwire Drive, at true scale. The cathode that left the Atacama as ore is now NM-B cable on a spool. Making one home's worth of it took roughly as much energy as that home uses in electricity in a month, and over its life, the wire will deliver several hundred times the energy it cost to make [dfw-c1 · validated](#). That asymmetry, cheap to make next to what flows through it, slow to replace, is the study's whole argument in one object.

4-5 · INSTALL & DISTRIBUTE

A distributor's warehouse, a contractor's van, a Tuesday. An electrician staples the wire into the stud bay of the house outside Dallas, and it disappears into the wall, where, if everything holds, no one will ever see it again.

Most supply-chain maps end here.

6 · DISPOSE

Ours doesn't; it turns around. The chain has three endings, and one of them isn't an ending: tailings at the mine, which began accumulating the day extraction started; insulation to landfill; and demolition scrap re-entering the smelting pool, a trade that rerouted twice in five years as import specifications tightened on both sides of the Pacific [scrap-g1 · validated](#). The dashed ring on the Malaysian node is honesty: we haven't named a facility yet, so the map won't pretend to a pin it hasn't earned.

There is a short loop as well as the long one. A tonne of clean scrap melted at the mill is ready to become rod again for roughly three to six percent of the energy it takes to bring a tonne from ore. That leaves out collecting, sorting and shipping the scrap, which nobody here has measured [swr-c1 · validated](#). The furnace's maker puts the ceiling at a fifth of the charge [swr-f2 · validated](#), and it is not the road demolition scrap takes: mixed and insulated scrap goes back through a smelter, and this study has not measured that road's energy at all.

Iron — the transformer

CHAPTER TWO · IRON

Follow one large power transformer, start to finish. Its paperwork would look something like this; the study fills in each line only as the referees validate it.

WHAT IT FEELS LIKE

The night the substation transformer dies, you hear it before you know it: the refrigerator stops mid-cycle. The utility's spare, if it has one, is at another substation, or it is a machine not yet built, at least three years out on a maker's order book. That machine's chain starts at the only mill in the country that makes its steel. `xfmr-f1 · validated; goes-f1 · validated`

THE MACHINE

The biggest of the machines this chapter follows weigh as much as a loaded 747

`move-f1 · validated; comparison cleared by the energy seat: "loaded" is load-bearing`, and every one is wound by hand. A worker guides copper wire, tens of tonnes of it in the big units (about forty in one maker's declared 500 MVA design) `xfmr-x1 · validated`, around a core of steel so specialized that sheets are rolled to nine-thousandths of an inch `goes-f2 · validated`. It takes months to build one, years to receive one, and when it works, you never see it: it sits behind a fence, humming, turning transmission voltage into the electricity your wall knows. The grid owns thousands of them `xfmr-f5 · validated: 4,900–6,799 LPTs`.

2 · THE STEEL

Transformer cores need grain-oriented electrical steel, GOES, and the United States has exactly one company that makes any: AK Steel, now **Cleveland-Cliffs**, at Butler, Pennsylvania `goes-f1 · validated`. It produces an intermediate grade American transformer builders mostly don't use; the top grade is made by one company on Earth: Nippon Steel, in Japan `goes-f1 · validated`. About 80% of the GOES in American transformers is imported `goes-f2 · validated`. And the one domestic mill is being crowded out of its own product: the same machines can make electric-vehicle motor steel, which pays better: the lithium chain competing with the iron chain for the same mill `goes-f2 · validated`.

3 · THE WORKS

A large power transformer is not a product; it is a project. Every unit is engineered to its substation: in the government's own words, "usually custom-built to fit the specifications of their use case"

`xfmr-f2b · validated`, and the heart of the build is still manual: winding crews, paper insulation placed by hand, master welders `xfmr-f6 · validated`. Intrinsic build time, with an open slot and no queue: about nine months in the last slack market `xfmr-c1 · validated, 2010 anchor`. The works on the map is a real one: Hitachi Energy's plant at South Boston, Virginia, building transformers since 1968, with three dated expansion steps between October 2022 and a groundbreaking in June 2026. The record gives dates and dollars and not one capacity figure, so no capacity is drawn from it `xfmr-f9 · validated`.

THE MOVE

Moving one is its own engineering discipline: 100 to 400 short tons, 200,000 to 800,000 pounds in the department's own words, `move-f1 · validated`, on one of a handful of special railcars in North America `move-f1 · validated`, with the last five to ten miles the hardest of the journey `move-f1 · validated`. The transport is itself a chokepoint: it binds exactly when many units must move at once.

IN SERVICE, AND THE AGING WAVE

The average American large power transformer is around forty years old, and seventy percent are twenty-five or older `xfmr-f5 · validated, DOE 2014`. Here the honest number matters more than the scary one: forty years is the *design* life, not a cliff. Fitted to the primaries, roughly four in five units sail past forty; the median unit lives nearly sixty years `xfmr-c1 · validated; hazard ESTIMATED bands stated`. The fleet is not about to fall over. It is about to need replacing at a rate the queue cannot casually absorb.

WHAT HAPPENED TO THE WAIT

Before the pandemic you could order a large transformer and receive it inside a year `xfmr-f1 · validated, DOE 2024`. Today, thirty-six months is the commonly quoted wait, sixty the recorded maximum `xfmr-f1 · validated`. In March 2025 GE Vernova's chief executive told an interviewer that the company's backlog of orders for gas turbines, power transformers and switchgear, taken together, stretched into 2028, and that he expected to be largely sold out through the end of that year by the summer `xfmr-f13 · validated`. Prices are up by about four-fifths on quoted prices, and by 63 to 68 percent on the federal transaction index, depending on the year `xfmr-f15 · validated, both bases`. Nothing was destroyed. Demand simply arrived faster than a bespoke, hand-wound, single-steel-supplier industry could grow `xfmr-f4 · validated`.

Our queue model puts the mechanical ceiling of the primary system at roughly **24–25 months**, even on the fastest order growth any maker has reported: the supply paths on record cannot produce the 36 the industry reports `e3-c1 · validated`. That gap is an adjudicated finding, not a mystery: the share of buyers paying to jump the queue is real but **small**: somewhere between one order in five and one in twenty, worth at most half a month at the median quote, not the missing twelve `sig-f1 · validated`. The residual belongs to three things this study cannot yet tell apart: how quotes themselves are formed, the un-primaried domestic half of supply, and the backlog the boom began with `seat ruling · unranked`. The pressure on the queue itself is on record: power plants and batteries holding signed or draft interconnection agreements add up to about seven and a half times what the country connected in 2024 `iq-f1 · validated`.

Every wait number in this chapter is a *floor*: the honest sentence is “at least this long” `referee ruling · binding`.

WHO WAITS

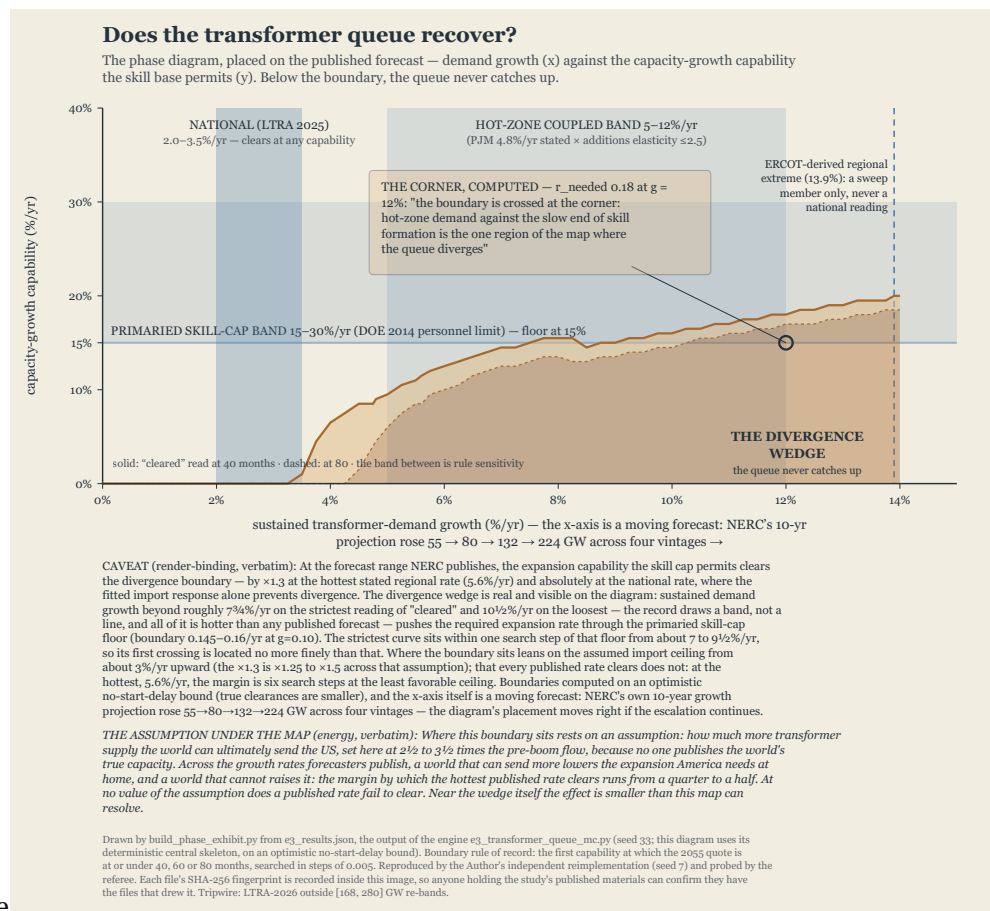
The engine's sharpest finding is distributional, and it renders as a mechanism, not a number: when buyers can pay to jump the queue, **the priority class waits about the build time (nine months, the build-time estimate the model is given, not something it computes) at every priority share we ran, while the standard class holds the entire queue** `e3-i1 · validated`. Record deliveries and record waits at the same time are not a paradox: a queue grows whenever orders outrun deliveries, however many are

delivered. What priority changes is who stands in it. Study 1 found that shocks bill the bottom of the income distribution; this chapter finds the same shape in industrial plumbing: *the queue is long, but it is not long for everyone.*

THE MAP'S QUESTION, ANSWERED

At what demand growth does a replacement queue stop being a wait and become a permanent state? **The boundary is computed, and the country's published forecasts sit on the safe side of it**

`xfmr-f7 · validated; exhibit cleared by the referee and the energy seat`: at the national forecast rate the queue cannot become permanent under any capacity response; the divergence wedge opens between roughly $7\frac{3}{4}$ and $10\frac{1}{2}$ percent a year, depending on what counts as cleared (the record draws a band, not a line), hotter than any published forecast. At the drawn edge, the wedge is crossed `corner sentence · fired verbatim`.



open full size

Both honesty notes render on the exhibit itself.

FRAGILITY, INSIDE THE GUARDRAIL

In April 2013, rifle fire damaged a number of transformers at this California substation; the one federal account declines to say how many, and so does this study `grid-f2 · validated, attributed`. Two things are true about that night, and the study insists on both: the grid *re-dispatched* (no blackout; redundancy did its job) and repairs to the equipment took nearly a month *with the power still on*. The lights never went out: the system re-dispatched around the damage. The hardware took nearly a month. Electrons reroute; iron does not. That gap is this chapter in one sentence.

Public record only; no method detail renders. The contested FERC power-flow claim renders in a separate evidence-tier exhibit, and nothing in our queue model speaks to how outages start no-bridge condition.

AGENCY: THE RESERVE QUESTION, MADE COMPUTABLE

In 2015 Congress asked whether America should keep a strategic reserve of transformers; the study concluded what the industry already knew, that bespoke machines make a warehouse of spares nearly useless, and the reserve was shelved for voluntary sharing programs grid-g1 · validated. Our engine turns that judgment into a curve: a reserve's value scales with **standardization**, the fraction of failures a stocked spare can serve, and that fraction is a *choice*, made a few hundred times a year by utilities each buying for its own system, none of whom is choosing it grid-g1 · validated. The buffer isn't missing because nobody thought of it. DOE's own account of why is regulatory: utilities may raise rates to recover from a disaster, and not to prevent one grid-g1 · validated.



The systems half of grid-g1, drawn static: as standardization rises a holder covers more of its own failures and a pool has less left to add; the anchored spare stock does not move; the record is quoted and the study takes no position. The tour carries the interactive edition.

Sand — the chip

CHAPTER THREE · SAND

Follow one power-management chip, the cheapest part in the charger, start to finish. Its paperwork would look something like this; the study fills in each line only as the referees validate it.

WHAT IT FEELS LIKE

You plug in the charger and nothing happens. The part that failed is a chip that manages power, cheaper than the cable, and the shop cannot get one either. When the shortage came, the median inventory of the chips buyers highlighted had fallen from forty days to under five. That chip’s chain starts in a North Carolina mine. fab-f2 · validated

THE CHARGER

The outlet from chapter one has a charger in it now. Inside the charger is the cheapest chip in the room: a power-management chip that turns wall current into the trickle a phone can drink. It costs less than a cup of coffee, it is made on factory lines that are decades old, and in 2021 the world ran out of it fab-f2 · validated. This chapter follows that chip backwards: to a furnace, a crucible, and a valley in North Carolina that a hurricane closed the same week it closed the valley next door, roughly fifteen to twenty kilometers away sand-x1 · validated.

1 · THE VALLEY

The crucible that holds molten silicon while a crystal is pulled from it is lined with quartz of a purity almost nothing else on Earth reaches, and most of it comes from one valley: Spruce Pine, North Carolina; two operators, Sibelco and The Quartz Corp spine-g1 · validated. How much of the world’s supply? **No agency publishes the share.** The United States Geological Survey carries no line for it at all; the 70-to-90 percent figures in circulation trace to industry-analyst estimates and to analyses repeating them spine-c1 · validated for the form. What can be said, attributed: “the only plant in the world right now that serves the semiconductor industry in its entirety,” in the words of the market-research chief who has tracked this chain for two decades. And what the careful counter-reading says, in its own voice: synthetic quartz exists at five to ten times the cost, China is scaling, the operators hold stocks: *a dominant source, not an irreplaceable linchpin*. The chapter keeps both sentences and lets you hold them.

Precision, kept honest: the quartz does not become the chip. It becomes the *inner layer of the crucible* that holds the melt spine-f2 · validated.

THE STORM AT THE MINE, AND THE INVERSION

On September 26, 2024 both operators halted. Power lines were the critical path. About two weeks later the restart was announced: production and shipments resuming, “progressively ramping up to full capacity.” How long the ramp took is not stated anywhere spine-f1 · validated; the mine’s clock is Sibelco’s: two operators halted, one restart is dated.

Here is the sentence this study was built to distrust, said by the operator six days after the storm: *“strong levels of feed stock in Norway... coupled with safety stocks of finished products and those that exist at different levels throughout the supply chain, we do not anticipate any critical situation.”* The other operator said it in fewer words: our final product stock has not been impacted spine-w1 · validated.

The most concentrated node in this study took a hundred-percent outage at its source and the chips never noticed, **because someone had kept slack**. The Quartz Corp says the decision was deliberate, taken after COVID; the feedstock sat in Norway because the firm is binational. How many weeks of cover that was, nobody has published. *Buffers bought the weeks; whether substitution could have bought the months was never tested.* spine-w1 · validated; spine-g1 · validated

The anti-just-in-time counterexample inside the just-in-time study: concentration survived here because the market around this node chose inventory over efficiency.

2 · THE CRUCIBLE AND THE MELT

The quartz goes to a crucible maker, a pattern until a plant is named cru-f1 · validated, where it becomes the inner lining of a vessel used once, or nearly once: the crucible degrades in the melt, the glass crystallizing at the 1,420°C melt face and dissolving slowly into the silicon, and cracks apart as it cools. Every pull eats a crucible.

The silicon itself arrives by the other path (quartzite, an arc furnace, the Siemens process) at about forty-nine kilowatt-hours per kilogram on the solar-grade basis the IEA publishes; the electronic-grade figure this chain actually runs on is higher, and nobody publishes it. Either way **the silicon in one chip cost about a watt-hour, a few seconds of a home’s electricity**. Where a chip’s energy really goes is the fab, and that number the chapter does not yet have poly-f1 · validated.

Who: polysilicon producers are chemical manufacturers and heavy users of electricity, as the sand chapter’s own figure for them shows. This study names none of them and has not sourced where their plants are. The work is process operation, maintenance and electrical trades. No headcount is given, and this study holds nothing on the communities beside these plants. A class, standing in until a plant is named. poly-g1 · validated

Who: the makers of fused-quartz crucibles are specialist firms. One is named in this study, at company level only; no plant is identified as making the crucible this chain follows. The work is furnace operation, forming and quality control. No headcount is given, and this study holds nothing on the communities beside these works. A class, standing in until a plant is named. cru-g1 · validated

3 · THE WAFER AND THE FAB

A crystal is pulled, sliced into wafers, a market of a small number of firms, with Japan’s SUMCO, by its own account, among the front-runners, and we have found no public agency that publishes the shares waf-f1 · validated, and the wafers go not to the famous five-nanometer fabs but to older, cheaper lines: the legacy-node factories where power-management chips are made, and where through the 2010s almost nobody added capacity.

When the pandemic’s demand wave hit those lines, the number that tells the story is federal: the Commerce Department’s supply-chain inquiry drew more than 150 responses and found that **the median inventory of the chips buyers highlighted had fallen from forty days in 2019 to less than five days in**

2021, naming, in its own words, “analog chips (used in power management...)” fab-f2 · validated. Five days. That is the just-in-time number of the entire chip shortage, and it belongs to this chip.

Who: the firms that pull silicon crystals and slice them into wafers. One, Japan's SUMCO, describes itself as among the front-runners; no public agency publishes the shares, and this study has not sourced where the works are. The work is crystal pulling, slicing and measurement. No headcount is given, and this study holds nothing on the communities beside these works. A class, standing in until a plant is named. waf-g1 · validated

CHIPS GOT PLANTS

What happened next is the part the next chapter will not be able to say. The legacy-node class *reorganized*: Texas Instruments broke ground at Sherman, Texas in 2022 on up to four 300-millimeter fabs, and in December 2024 signed for up to \$1.6 billion of federal CHIPS funding across three fab-f1 · validated; fab-g1 · validated. Announcement to first wafer runs about three years, the same shape as the transformer chapter’s expansion delay.

That the class stagnated *before* 2020 is the chapter’s claim, not yet its primaried fact: the trade body’s capacity numbers are forecasts and the chapter labels them as such fab-f1 · validated.

6 · THE PARTIAL RETURN

The copper chain came back around as a cycle. This one does not, quite: when the charger dies, the board goes to an e-waste stream where the copper and gold are recovered and the silicon is lost, work done by a recovery workforce, formal and informal, in the places that do the world’s recovering

ew-f1 · validated; ew-g1 · validated. The copper on the board re-enters the copper chapter’s scrap pool (the map draws that arc) sand-x4 · validated, and the chip itself ends. Three chains, three endings: copper cycles, silicon ends, and salt, next, mostly burns.

Salt — the IV bag

CHAPTER FOUR · SALT

Follow one bag of intravenous saline, start to finish. Its paperwork would look something like this; the study fills in each line only as the referees validate it.

WHAT IT FEELS LIKE

You watch the drip. The bag on the pole is salt water, the plainest thing in the hospital, and one flooded plant in North Carolina made about sixty percent of what American hospitals use in a day, on the hospital association's figure. Within days, supplies were being reserved for certain patients. The chain begins in a salt dome. nc-c1 · validated; nc-w1 · validated

THE BAG

A liter of salt water in a plastic bag, hanging from a pole. About 1.5 million of them are used in American hospitals every day from one plant alone: “approximately 60% — or 1.5 million bags,” in the hospital association's words, the “approximately” governing both nc-c1 · validated, attributed. Two kinds of salt meet in it: the sodium chloride dissolved in the water, and the chlorine that became the bag.

1-2 · THE BRINE AND THE CHLORINE ROOM

America produces about forty million tonnes of salt a year, and the cheapest form is brine pumped from domes along the Gulf Coast: about ten dollars a tonne, against fifty-six for rock salt brine-f1 · validated. Roughly fourteen million tonnes a year flow to the chlor-alkali belt. The survey names no operations, so the map draws a ring, not a pin, and the Who stays at that precision too: the dome operators, and the parishes and counties that sit over the domes brine-g1 · validated.

Electrolysis splits brine into chlorine, caustic soda and hydrogen: one tonne of chlorine, 1.1 tonnes of caustic, 30 kilograms of hydrogen, always, in that ratio chl-f1 · validated. It costs about 2,600 to 2,900 kilowatt-hours per tonne of chlorine, *roughly three months of a household's electricity for every tonne*; the European industry alone draws 20 to 26 terawatt-hours a year, the electricity of 1.9 to 2.4 million American homes chl-c1 · validated. **The lock:** chlorine and caustic are made together and sold apart, and chlorine cannot wait (it liquefies badly and stores worse) while caustic sits happily in tanks. So production follows chlorine, and caustic floods or starves chl-f1 · validated.

2B · THE VINYL WORKS, AND WHO IS BESIDE THEM

Where the chlorine goes next is a stage this chapter draws as a ring and has not stopped at: the plants that turn it into vinyl, and the people beside them. This study names one operating vinyls plant, at Plaquemine, and one proposed for St. James Parish whose permits were vacated in 2022, reinstated in January 2024 and left standing when further review was refused that September; it is not built, and the case is described by where it stands, never as settled. The people along these reaches appear here through the public record only: that litigation, and the property-line monitoring these plants were to begin in July 2026, deferred for

two years by a proclamation filed the morning it took effect `chl-g1 · validated`. And one community in this belt no longer exists as a place: Mossville, in Calcasieu Parish, founded over two centuries ago by free Black families and bought out under a voluntary purchase program from 2013. That buyout was Sasol's, whose plants there are not this chain's; it is here as the limit case of the question this study keeps asking, what passes and what redraws the map, and not as this bag's doing `pvc-g1 · validated`.

3 · THE BAG WORKS, AND THE STORM

Marion, North Carolina: ten manufacturing lines, more than 2,500 employees, and the about-1.5-million bags a day. Helene's rain came down the valley on the morning of September 27 `helene-f1 · validated`; by September 29 the plant was "affected by flooding and currently closed for production." The clock that follows has three marks, and the chapter names each one: **the first line**, the highest-throughput one, a quarter of the plant's output, back on October 31, thirty-two days in; **all ten lines** on January 28, one hundred and twenty-one days in; **pre-hurricane output** on February 17, day one hundred and forty-one `nc-f1 · validated`.

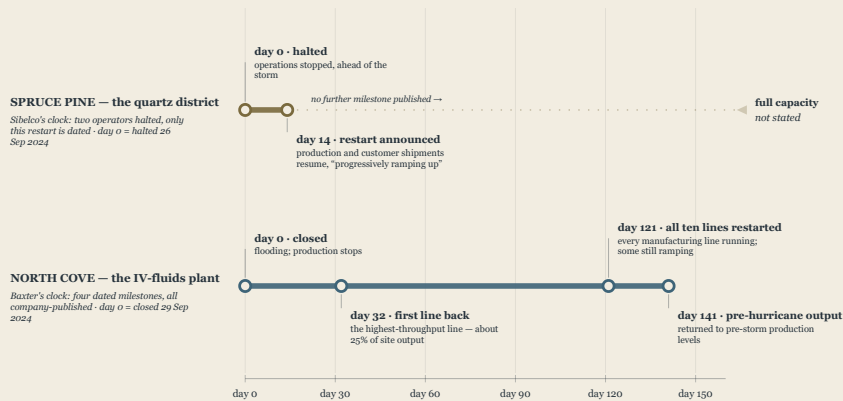
One honesty note rides with the storm: the system was already short before it. Of the nine IV products on the FDA's shortage list after the storm, six had been listed before it, the common dextrose solutions since February 2022 and normal saline since June 2018; three were new. The storm mostly deepened listings that already existed `nc-f1 · validated; nc-f2 · validated`.

THE TWO VALLEYS

Same storm. Roughly fifteen to twenty kilometers apart. Closures three days apart. And then two clocks that do not line up, so the chapter draws the whole ladder rather than a single bar. Matched first production to first production, the plant took a bit over twice as long as the mine. Matched restart-to-all-units, the gap is nearer nine times. And the comparison everyone reaches for, two weeks against "fully recovered," **cannot be made at all**, because the mine never published a full-capacity date `sand-x1 · validated; salt-x1 · validated`.

One storm, two clocks

Hurricane Helene closed the world's dominant high-purity quartz district and the plant that makes most of America's IV fluids in the same week. Each site's clock starts at its own closure; every mark is labeled for what it is; and a milestone the record never published is drawn as an absence rather than left out.



WHAT MAY BE COMPARED, AND WHAT MAY NOT

first production → first production — day 14 against day 32 ($\times 2.3$): the matched pair — the plant took a bit over twice as long
 restart announced → all units running — day 14 against day 121 ($\times 8.6$): a wider, unmatched pairing, stated as such
 full output → full output — CANNOT BE FORMED — the mine never published a full-capacity date

Roughly fifteen to twenty kilometers apart — a range, not a bound, because it depends what you measure: 15.1 km and 17.4 km between mapped footprint centroids, 21.4 km between the registers' district-precision map pins, less than any of these footprint-edge to footprint-edge; the road distance is longer. Corrected from an earlier "thirty-five kilometers" on the referee's finding, stated as a range, showing the measure that breaks the bound.

What separates a two-week node from a four-month node is not the storm: it is depth of damage times the regime that governs restart — a mine restarts when the power line is repaired; a sterile-pharmaceutical plant restarts when every line is requalified. On the felt-impact side: where the buffers sat.

Common driver, not contagion: one storm struck both chains independently — neither failure propagated to the other.

Concentration claims above are the conservative form of claims attributed in full in the chapters: "dominant" is the skeptical counter-analysis's own concession word (no agency publishes a share — see spine-c1, Render Form A), and "most of America's IV fluids" is softer than the "approximately 60%" the hospital association states (nc-c1).

Drawn by build_helene_ladder.py. Its dates and quotations are written into that file, each from a published document: Sibeco's, company releases on the halt (2024) and the restart (30 Oct 2024) at Spruce Pine; Baxter, a customer letter hosted by the FDA (30 Sep 2024) and Hurricane Helene updates (31 Oct 2024, 28 Jan 2025, 17 Feb 2025). The generator's SHA-256 fingerprint, and the address, fingerprint and retrieval date of each document, are recorded inside this image, so anyone can confirm they hold the file that drew it and the documents it was drawn from. Register rows sind-v1 / salt-v1 (validated). Endpoint labeling, pairings and the distance range are seat railings.

open full size

Cleared by two seats and re-confirmed after a layout rebuild; built by an auditable generator with five build gates.

What separates a two-week node from a four-month node is not the storm. It is **depth of damage times the regime that governs restart**: a mine restarts when the power line is repaired; a sterile-pharmaceutical plant restarts when every line is requalified, and on the felt-impact side, *where the buffers sat*. The sand chain had stock in Norway. The salt chain began rationing within days of the closure.

THE SHORTAGE AT THE BEDSIDE

Nearly eighty-five percent of four hundred hospital pharmacists surveyed in mid-October reported a moderate or critical impact, critical meaning treatments delayed or canceled; they converted infusions to injections, bought premixed drugs, used smaller bags, reserved supplies for certain patients. In the same survey, seventeen percent of all participants reported canceling or delaying voluntary surgeries

`nc-w1 · validated, ASHP, attributed`.

FDA authorized imports from nine sites abroad, three weeks in, and by year's end an airlift of nearly two hundred flights had moved nearly eighteen thousand tons of product `nc-g2 · validated`. And FDA said, in its own words, what it could not do: "FDA does not have authority over allocation or distribution of these products" `nc-g1 · validated`.

THE THIRD CLOCK

The plant was back in about four months. The shortage was not over until August 2025: “approximately eleven months,” in the hospital association’s statement of record to the Senate. The seven months in between are the pipeline refilling and the allocations unwinding, and they are this chapter’s sentence: **the system recovers when the buffers refill, not when the plant restarts** nc-f1 · validated.

THE REPEAT OFFENDER

In 2017 Hurricane Maria hit the same company’s plants in Puerto Rico and the country ran short of saline; the FDA Commissioner said the shortages “resulted from, or were worsened by” the storm. The shortage listing for normal saline that FDA posted on June 21, 2018 **was still open when Helene hit**: one unbroken listing, deepened by two hurricanes seven years apart nc-f2 · validated. The response was the same playbook both times (emergency imports, new entrants, extended expiry), faster in 2024. **The response layer reorganized. The capacity layer did not.** The bridge got better; the plant count did not.

Chips got plants. Bags got airlifts. Whether a shock de-concentrates a chain depends on the market structure around the node, not on the shock.

LEARNED TO RESPOND, NOT TO BE LESS FRAGILE

Between Maria and Helene, American institutions learned to respond: the same FDA toolkit, used faster, with an airlift and a Defense Production Act order for one remediation material. They did not learn to be less fragile. The ledger, all on the primary record: the law requires firms to keep a “redundancy risk management plan,” kept by the firm, inspected only, “as appropriate”; a proposal to pay hospitals for buffer stock was solicited and “not adopted”; the strategic reserve “focuses on... countermeasures that do not have a commercial market”; a six-month reserve bill sat unenacted in October 2025; and the 2024 federal white paper on preventing shortages never mentions Maria, Baxter, saline, or IV fluids nc-g2 · validated.

The transformer chapter found a reserve question with an owners’ club to answer it. This one is its medical twin, with the club missing.

Oil — the tank of every truck

CHAPTER FIVE · OIL

Follow one tank of diesel, start to finish: the leg underneath every other chain in this study. Its paperwork would look something like this; the study fills in each line only as the referees validate it.

WHAT IT FEELS LIKE

You find the shelf empty, then the pump bagged, then the ward waiting on a delivery that did not come, because the last leg of every other chain is a truck. Mid-August 2026 the country held about 27 to 29 days of distillate cover, a ceiling on comfort and not a countdown. The chain starts in a Texas oilfield. oil-b2 · validated

THE FUEL

Four chains have ended in this house. Every one of them arrived on a truck.

This chapter follows the fuel in that truck, diesel, which is the only product in the study whose map is **every other map's stage five**. It is not a fifth chain beside the others. It is the leg underneath them.

1 · EXTRACT

The crude starts in west Texas and southeastern New Mexico. Two states produced **58.9%** of American crude oil in 2025 oil-c1 · validated.

The number that matters is not that one. It is the shape behind it: that share **fell** for two decades, to 22.6% in 2002, and then rose **2.61×**. Concentration here is not the resting state of the system and it is not fate. It was reduced, and then it was rebuilt, inside one working lifetime oil-c1 · validated.

The pin is a ring, not a dot, **because the evidence is state-level**: the record names two states, not a wellhead, and the map says only what the record says. Read that as a limit of the record and not a description of the place: the people who live above this rock are not absent from it, they are absent from a figure drawn at the wrong scale to hold them seat ruling.

WHO OWNS THE ROCK

Ownership here is not one thing, and the part of it that can be traced leads somewhere unexpected. In New Mexico, some of this rock is **state trust land**, and the royalties have a named destination: the **Land Grant Permanent Fund**, set up when the state entered the union in 1912. It pays out **more than a billion dollars a year** (\$1.34 billion in 2024) to public schools, universities and other beneficiaries, and the money reaches it *monthly* who-up2 · validated.

The regulator is stranger still. Texas oil and gas is overseen by the **Railroad Commission of Texas**, created in 1891 to stop railroads overcharging farmers. It has not regulated a railroad since **2005**, when the last rail duties went to the state highway department. The name is a fossil the institution never shed

who-up2 · validated.

Two things this beat does not say. It does not say how much of *this* oil pays that rent: the split between state trust, federal and private land is not sourced, so the fund is context and not attribution. And it puts no number on the people who work here: this study holds no count of them, so the headcount is left out on purpose.

WHAT THE REGULATOR HAS HAD TO DO

Oil does not come up alone. It comes up with water, and the water is injected back underground, which is where a second story starts, because the injection has been making earthquakes.

The Railroad Commission's answer was to draw **Seismic Response Areas** on the basin and act inside them. In the Northern Culberson-Reeves area it **suspended every deep disposal permit**, 23 wells, effective January 12, 2024. Around Stanton it cut daily injection volumes and told individual wells to stop

`who-up3 · validated`.

Two things in that record are worth more than the enforcement. The first is a **clock**: in the Culberson-Reeves area the Commission estimates **twelve to eighteen months** between injection rising and seismicity rising: its own estimate, not a measured interval. The ground answers a year to a year and a half late, which means the cause is always older than the alarm. **That figure is for injection going up.** Asked how long a decrease takes to show, the Commission calls it uncertain in the same breath, and publishes no number at all. The physics says which way that uncertainty runs: shutting a well stops *adding* pressure, but it does not take back the pressure already in the rock, which keeps spreading for months before it decays. **The wait for a fall is longer than the wait for a rise**, so every number below is measured early, with some of the decline still arriving `who-up3 · validated; seis-x1 · validated`.

The second is that the Commission says out loud where it sets its own buffer. To hold off a magnitude-5 event it acts when quakes are running between **magnitude 3 and 4. It acts at three to prevent five.** That is a buffer decision, written down by an institution, in the middle of a chapter about buffers.

And the instrument changed under them. In January 2022 the Commission asked existing wells for what existing wells can give: six months of daily volume and *surface* pressure. Thirteen months later it began requiring new wells to carry a **bottomhole pressure gauge** and report both pressures monthly `who-up3 · validated`. You can watch a regulator learn what it needs to measure by reading two of its own documents a year apart.

The response plan had set a *goal*: that magnitude-3.5 quakes be decreasing by the end of 2023. So: did any of it work? The record cannot say, and *how* it cannot say is the useful part.

Counts of those quakes did fall inside the response areas: by about half around Culberson-Reeves, about three-fifths around Midland. But in a nearby part of the same basin with **no response area at all**, they fell by about seventy per cent. The treated ground improved; the untreated ground improved more. And Culberson-Reeves had already been falling for a year before the suspension was signed `seis-x1 · validated`.

Whether the *biggest* quakes responded is a question the record cannot answer, and it is worth seeing why. Magnitude-5 events run about three-quarters of one a year here. To detect a halving at that rate you would need roughly sixty years of before and sixty years of after. There are four, and under three. The largest quake before the suspension was magnitude 5.4; the largest after it was **magnitude 5.4, on May 4, 2025**. Both are public record. Neither is evidence of anything `seis-x1 · validated`.

That is the goal's real problem, and it is sharper than a drafting error. A goal written on the *number* of magnitude-3.5 quakes can be checked in a few years. A goal written on the *largest* event cannot be checked for about a century. **The Commission set its goal on the indicator it could verify, and then acted on the indicator it could not.** That is not special to Texas. When a hazard is heavy-tailed and the policy runs on a policy timescale, the thing you can check and the thing you need to know are different objects.

So far this looks like a question the record cannot answer. It turns out that it can, but not by asking where. The areas are drawn approximately here.

THE WRONG CONTROL

Twenty-three wells stopped taking water. The water did not stop being produced, and for a while the obvious question was where it went, because quakes in a ring outside the response area rose while quakes inside it fell.

That turned out to be the wrong question, and the reason is worth more than the answer. **The Commission did not close disposal in an area. It closed deep disposal.** The order is named for a place, but the thing it actually selected on was *depth*, and a comparison built on the map was testing the label instead of the mechanism.

The right comparison was inside the treated area the whole time: the shallower rock underneath the same ground, with the same operators, the same instruments and the same regional weather of activity. Split that way, the deep earthquakes nearly stop, down about **92%**, while the shallow ones fall by about **48%**, roughly what the whole basin was doing anyway seis-x1 · validated.

So the record *can* partly separate the order from the trend. Not by where the quakes were. By how deep they were.

Held loosely, and the reasons are stated because they bound it: the depth of an earthquake is not the depth of a disposal well, the catalog resolves depth to about a kilometer, the deep counts are small, and the whole thing is measured early. The ring outside the area did rise, mostly in New Mexico, but if the deep water had crossed the state line the new quakes there should be deep, and they are not: about one in eight. New Mexico was growing on its own, and the tidier story, that a closed gate simply moves the flow next door, was tested and did not survive.

That test has since been run, at least in part, and the answer is no. If the suspended water crossed the line, somebody had to open somewhere to put it. Nobody did. New deep disposal wells on the New Mexico side peaked at **thirty-one a year** in 2018 and 2019 and were down to **five a year by 2021**, a year before Texas wrote its first response plan. Across the four years before the interventions: eighty new deep wells. Across the four years after: **twenty-one**. Not one intervention is followed by an increase seis-x1 · validated.

So the explanation holds together now without displacement in it. New Mexico's new earthquakes are shallow because New Mexico's new injection is shallow, and it is New Mexico's own. Two instruments, arrived at separately, one story.

One door is still open, and it is worth naming exactly rather than gesturing at. This tested new capacity: whether anywhere was built to take the water. It did not test the volumes going into wells that already existed. When we went looking for those, they were not there: New Mexico's published disposal-well records carry *no volume figures at all*, for any of the 1,066 wells in them. That channel is untested, which is not the same as tested and clear seis-x1 · validated.

THE OTHER REGULATOR

There is one more reason the water-crossed-the-line story does not hold, and it is not about geology at all. **New Mexico was never a bystander.** It was already regulating induced earthquakes of its own, and by a different theory of what causes them who-up3 · validated.

Texas asks *which rock you inject into*: the suspension closed deep disposal, by formation. New Mexico asks *how close you are to what already happened*. Two magnitude-2.5 quakes inside thirty days and ten miles, and monitoring starts. One magnitude-3.0, and injection rates come down by distance from it. A magnitude-3.5 within ten miles, and wells within three miles **shut in**.

The dates are the part worth having. Texas published its first response-area map on **October 21, 2021**. New Mexico's protocol is dated **November 23**, its public map layers **December 20**, and Texas's statement of guidance **January 31, 2022**. Four regulatory acts, two agencies, two states, inside fourteen weeks.

So neither state was following the other. What that looks like is **one professional response, seen from two sides of a line**, which is a different thing from a hazard migrating, and a better explanation of why both maps appeared at once.

This is *not* a fifth case of the pattern in "How we know any of this." Two state oil-and-gas regulators, one hazard, one basin split by a survey line, the same year: that is one observation seen twice, not two. The count stays at four seat ruling.

2 · PROCESS

Crude is not fuel. It becomes diesel on the Gulf Coast, and the eight largest American refineries include seven in one region. The first one outside it is eighth, in Indiana oil-n1 · validated.

Port Arthur, Texas alone holds three refineries, together 7% of national capacity. The largest single refinery in the country is one of them.

Why *there* is a question this chapter does not answer. Three explanations are available and a capacity table cannot separate them: the geology put the oil nearby, the ports made it cheap to move, and the permitting made it hard to build anywhere else. They may not even compete: the likeliest reading is that they ran in sequence, and that what looks like a choice is a position inherited and then frozen.

Two chapters now leave the same coastline: the salt chapter opened on Gulf Coast salt domes, and this refinery stands on the Gulf Coast. That is not a coincidence to make anything of: it is what a coastline with ports, pipe and salt domes attracts.

WHO LIVES WITH IT

Three refineries in one town, and three very different owners. EIA's own capacity report names each parent: Motiva is **wholly owned by Saudi Aramco**, the largest of the three; the second files under Premcor and belongs to **Valero**; the third is **TotalEnergies** who-rf1 · validated. Seven per cent of American refining capacity, in one town, held by a foreign state oil company, an American independent and a French major. The capacity figure was a fact about throughput; the ownership makes it a fact about governance.

The town is **Port Arthur, Texas**: 56,039 people at the 2020 census, in a county of 256,526. Median household income is **\$45,752** against **\$76,292** for Texas (three-fifths of the state figure), and the poverty rate is **29.1%** against the state's 13.8% who-rf2 · validated.

The most documented environmental record here is *not* about the refineries, and getting that wrong would name the wrong facility. In 2021 the **Port Arthur Community Action Network**, with Lone Star Legal Aid and the Environmental Integrity Project, filed a civil rights complaint, not against a plant, but against the **state regulator**, over its oversight of a petroleum-coke facility that calcines the refineries' own byproduct. The complaint is that the emissions were *inadequately monitored or quantified*, and so not properly weighed when permits were decided who-rf5 · validated. EPA investigated for two years, the state withdrew from negotiations, and the file was closed without a finding either way: re-filable, and the agency said it would keep gathering data regardless. Not resolved, not rejected, not live.

And the neighborhood that complaint names is **West Port Arthur**. This study puts no number beside that name, and the reason is worth more than a number would be: **West Port Arthur has no boundary anyone tabulates**. The census counts cities and tracts; the harm has a neighborhood. Any figure we printed would be a line we had drawn ourselves, wearing the Census's authority seat ruling. The geography the harm has and the geography the state counts are not the same geography, and here the instrument that does not fit is the census itself.

The 56,039 above describes the town that hosts the complex. It is not a count of anyone's exposure to anything who-rf2 · validated.

THE SLACK QUESTION

Every chapter so far has asked how deep the buffer is. In refining, the buffer has a name: **idle capacity**. A plant running at 91% has 9% of itself in reserve; a plant running at 96% has almost none.

The obvious story is that the slack was squeezed out. That story is wrong, and it was the first thing I wrote down, because it is what anyone expects oil-c2 · validated. Utilization climbed from 77.6% in 1985 to a peak of **95.6% in 1998**, and today is 91.9%, *below* that peak. The tightening is an era, not a trend.

WHY IT MOVED

Utilization is throughput divided by capacity, so the change splits exactly into the two oil-c3 · validated. Which window you choose decides the answer, so both are here, each with the question it actually answers.

What brought utilization back after 2020? Demand. Throughput rose 13.4% while capacity fell 2.6%: 83% of the movement is the numerator oil-c3 · validated.

Is the system tighter than before the pandemic? Yes, and not because it is busier. Since 2019 throughput has fallen 1.7% and capacity 3.4%: the plant runs hotter on *less work done*. Capacity peaked at 18,808 thousand barrels a day in 2019, bottomed at 17,963 in 2022, and has clawed back about a quarter of that, not steadily, and still 635 below its 2019 peak oil-c3 · validated.

Slack in this system is a residual, never a reserve, and when it persists, it is removed rather than carried.

5 · DISTRIBUTE

Downstream of the refinery the buffer stops being idle steel and becomes something simpler: tanks. The question a reader can hold is **how many days of it there are**.

Mid-August 2026: about **27 to 29 days** of distillate cover: stocks of *distillate* against *distillate product supplied*, averaged over *four, thirteen and fifty-two weeks*, against about **32 to 34** on the same week in 1996, measured the same way [oil-b2 · validated](#): **four to five days less**, and the lowest of any comparable August in the record. That difference is taken basis by basis, not by subtracting the two ranges.

Read that as a ceiling, not a countdown. Tank heels and pipeline linefill sit inside the number and are available to nobody; exports are netted out; and it is an August rate standing against a winter risk. All three point the same way: the real buffer is *shorter* than the figure.

THE LEG UNDERNEATH

Those three lit lines are the final leg of three of the other chains: the wire to the house, the transformer to the substation, the saline to the ward. Every one of them is a truck, and every one of those trucks was running on the product this chapter just built. The chip's last leg is not on this map: the study does not locate where it starts.

The other chains have chokepoints of their own. This one is different in kind: it does not have to break to be felt everywhere, because it is already inside every other chain's last mile.

WHAT THE FUEL CHAIN IS

Diesel is the only product in this study that is not a thing you own. Nobody's house contains a tank of it. It is the one that moves everything you do own.

A severed diesel link is felt two ways, and the Roundtable has now measured one each. **The price channel** was Study 1: a median **\$23.3 billion a month** in added household cost, falling hardest where there is least: 8.9% of *after-tax* income at the bottom against 1.6% at the top, regressive in every one of a hundred thousand bootstrap replicates [oil-s1 · validated](#). **The physical channel** is this chapter: not what it costs, but how many days of it there are.

Neither stands in for the other, and it matters that we say so. Study 1 holds no freight layer: it priced a shock, it did not delay a delivery. This chapter holds no price model. The honest statement is that one severed link has two clocks, and we have wound each of them separately.

This chapter deliberately stops at the terminal. What a disruption would feel like day to day is a question the closing chapter takes up, on evidence, for all six materials, and it is not a question the Author gets to answer alone.

Lithium — the grid battery

CHAPTER SIX · LITHIUM

Follow one grid battery, start to finish. Its paperwork would look something like this; the study fills in each line only as the referees validate it.

WHAT IT FEELS LIKE

Seven in the evening, the hour the grid's batteries were meant to carry, and your lights are running on something else. The cell that did not arrive cannot be hurried: brine spends about eighteen to twenty-four months in Albemarle's ponds, by its own account, and longer if the weather is poor. The chain starts in the driest desert on Earth. li-e1 · validated

A LAKE YOU CANNOT DRINK

High in the driest desert on Earth, there is a flat white plain the size of a small country. Under its crust sits water, but water with so much salt dissolved in it that it is closer to syrup than to anything you would pour. Pumped up, it is not a drink and not a mineral. It is a solution, and the thing worth having is dissolved in it.

Everything in this chapter follows from that one fact. The other five chains in this study begin with a rock that is dug. This one begins with a liquid that is pumped, and a liquid behaves differently. It moves, it refills, and it is shared with whatever else is drawing on the same ground.

Almost every pin on this chain is a ring, not a dot. A ring means the study is showing you a stage, not naming a plant. **One of the six has earned a dot**, and it earned it after this chapter was drawn, which is the honest state of a chapter this young.

ONE AQUIFER, THREE COMPANIES

The salt flat has a neighbor. Copper, chapter one of this study, the wire behind the outlet, is mined a short drive away, and it too needs water.

In December 2024 a Chilean environmental tribunal approved a settlement that named them together. A copper mine, a second copper operation and a lithium producer were sued jointly by the Comunidad Indígena Atacameña de Peine over damage to one named body of groundwater. The agreement binds all three to fourteen measures worth about US\$47 million, the first of which is that aquifer's recovery, and the sixth of which is a standing table where the community and the companies sit together li-x1 · validated. And the tribunal approved it on a **divided bench**: the dissenting minister's ground was that no measure in the agreement repairs the aquifer drawdown itself, recovery estimated at around the year 2200, and only under an immediate cessation nobody signed. Implementation runs under the tribunal's continued supervision, and unevenly: the standing table functions, while the baseline study that most of the measures wait on is still at terms of reference esc-g1 · validated.

Two chains in this study are linked, and it is worth being exact about how. They are linked by an *instrument*: one community, one proceeding, one shared bill. They are *not* shown here to be linked physically: whether pumping brine and pumping fresh water act on the same water is disputed among the

people who study it, and this study does not settle it li-x1 · validated.

This link had to be earned. Before anyone looked, the question was written down: *if the lithium producer turns out not to be a party, there is no link*, so that the answer could not be arranged afterwards seat rule.

THE CLOCK MONEY CANNOT SHORTEN

To get lithium out of that brine, the standard method is to let the sun do it. The solution is moved through a series of shallow ponds and left there while the desert takes the water off. What stays behind grows steadily more concentrated, pond by pond, until it is worth trucking to a plant that turns it into a white powder.

This is the first clock in the study that money cannot shorten, and the reason is the interesting part. The transformer queue in chapter two is long because there is not enough plant and not enough skill: build more of both and the wait comes down. Evaporation is not like that. Build more ponds and you get more tonnes; you do not get them sooner. Every tonne still has to sit in the sun for its own turn seat ruling.

How long is that turn? Longer than the figure in circulation. Albemarle, describing its own ponds on this salt flat in its own annual report: **about eighteen to twenty-four months**, and *“if weather conditions are not favorable, the evaporation process may be prolonged”* li-e1 · validated.

That last clause is not decoration. **It means twenty-four months is not a ceiling:** the company declines to put one there, and neither will we.

Note what it names as the thing that stretches the clock. Not plant, not people, not money. **The weather.** That is what a residence time is, said by the operator rather than by us.

The number usually given is a year to eighteen months. This study declined to print it while it was unsourced, and the operator's filed figure turns out to *begin* where that one ends.

One company's ponds, not the salt flat's, and that is a fact about the record, not a doubt about the number. **The other operator on this salar describes the same pond sequence in more detail than Albemarle does, and never says how long it takes.**

Who: Albemarle runs the carbonate plant, and it is also one of the companies in the Peine settlement at the salt flat. The company links the two places; the community does not. Peine's case is about groundwater at the salt flat, not about this plant. Around the plant is an industrial district of the Antofagasta region with people living in it, and this study can say no more about them than that: it holds no Chilean regulatory record for the plant itself. lng-g1 · validated

MINED IS NOT REFINED

Ask who controls lithium and you will be given a number. Which number you get depends on a distinction almost nobody draws.

Digging it up is one industry. Turning it into something a battery can use is another, and they do not happen in the same places. At the mine stage no one dominates: Australia leads with rather more than a third, Chile about a fifth, China about a sixth, and the top three together come to roughly three-quarters, shares of a world total in which the United States figure is **withheld**: USGS reports American mine production only as “W”, so every share here divides by a total containing a withheld, non-zero producer li-c1 · validated.

At the refining stage the picture changes completely. On the same year's reading, on the IEA's published chart, China holds around seventy per cent of the world's refined lithium, and the top three between them hold nearly all of it [li-c2 · validated](#).

The same material, the same year, two different worlds, and the more quotable figure is the one that answers the question nobody asked. The most-cited source in the field publishes mine production and reserves, and does not report refining at all [li-c1 · validated](#).

Who: nobody, and the reason is worth a sentence. This stage is a world total, not a place, and adding up is the operation that removes people. Five producing countries stand behind the figure, with five workforces, five sets of neighbors and five regulators; the number sums their tonnes and nothing else. Where the chain reaches a place again, the people return.

[mine-g1 · validated](#)

AGAINST COPPER, ON ONE CHART

Concentration is only alarming next to something. So here is the comparison this study can actually make honestly: the same chart, the same year, the same stage, the same way of counting.

Refined lithium: about seventy per cent in one country, and the top three at nearly all of it. Refined copper: about forty-four per cent in that same country, top three about sixty. Both chains are concentrated. One is markedly more so [li-c2 · validated](#).

And the same chart is why this chapter will not call lithium the most concentrated of anything. Read across it: battery-grade graphite sits near ninety-five per cent in one country and magnet rare earths near ninety-one, both above lithium. Against copper, the one other of this study's six that the chart covers, lithium's refining is markedly the more concentrated. Among critical minerals generally, it is not close to the top [li-c2 · validated](#).

These shares were read off a printed bar chart at high magnification, good to a point or two, which is why they are written here as "about". They are a reading, not a measurement, and the register says so on the row.

WHAT THE POWDER IS MADE OF

Lithium on its own stores nothing. It has to be built into a cathode, the powder that does the actual holding, and there is more than one recipe. Which recipe a battery uses changes what else has to be mined for it, how long it lasts, and how much of it you need.

That opens a door this chapter is deliberately not walking through yet. One of the common recipes for grid storage uses iron. If that is the recipe here, this chain reaches back into chapter two's material, and the study has a link it did not plan for.

It is not claimed, because it arrived by looking attractive. A shape that suggests itself before it has been tested has to say in advance what would prove it wrong. So: the question is whether the iron in such a cathode passes through any part of the chain the iron chapter actually follows. If it does not, if it is a different grade, a different supplier and a different route that merely share an element, **there is no link**, and the honest answer is two chains that use the same word [li-p2 · ruled: not built](#).

Who: the makers of cathode powder are specialty chemical producers. This study names none of them and has not sourced where their plants are. The work is preparing the precursor, firing it in kilns, and quality control. No headcount is given, and this study holds nothing on the communities beside these plants. A class, standing in until a plant is named.

cam-g1 · validated

Who: the makers of lithium-ion cells. This study names none of them and has not sourced where their plants are. The work is coating electrodes, assembling cells, and the slow first charging and testing that follows. No headcount is given, and this study holds nothing on the communities beside these plants. A class, standing in until a plant is named. cell-g1 · validated

THE THING THAT KEEPS THE LIGHTS ON

Cells are stacked into modules, modules into packs, packs into containers about the size of a shipping crate, and a row of those containers is wired into a substation. That is where this chain ends: not in your pocket, but in a fenced yard somewhere near you, holding a few hours of the grid.

That was a decision, and it was the Editor's. A phone battery was the obvious pick and the wrong one: a person keeps a phone for years, and one phone dying is not a systemic event. This study is about failures that are felt beyond the person they happen to.

Which lands this chapter beside chapter two, arriving from the opposite direction. A transformer and a grid battery are two answers to the same question: what stands between the generator and the house. One got there through iron and a queue years long. The other through a salt lake and an ocean. *That is a resemblance, not an argument:* it describes two findings each established on its own evidence, and it is offered as support for neither.

Who: the firms that build cells into modules, packs and containers. Where that is done is not established in general. In the one case this study has sourced, the maker's own annual filing names dedicated factories for the product in Lathrop, California and in Shanghai: neither where the cells are made nor where the battery is installed. Other makers may do it differently. The work is module assembly, enclosures, cooling and commissioning. No headcount is given. This study has not looked for a community record at either factory, so it states none. pack-g1 · validated

WHAT THIS CHAPTER DOES NOT HOLD

This chapter stands on more than it did when this ledger was first written, and the ledger says so rather than freezing. Here is the honest state of it.

Validated: the shared-aquifer settlement li-x1 · validated, how long the ponds take (the operator's own filed figure li-e1 · validated), the mine-stage shares li-c1 · validated, where this product's modules and packs are actually assembled, in the maker's own annual filing li-p1 · validated, and the refining shares with their copper comparison, as a reading off one published chart and scoped to it li-c2 · validated.

Closed without a link: whether the cathode reaches back into iron. The test was never run: it rested on a premise the study could not source, that the iron recipe is the one in use here, and a link that cannot be sourced is not built li-p2 · ruled: not built. That is not a finding that the two chains are separate.

One thing the clock does tell us, and it is the same number twice. Every tonne spends eighteen to twenty-four months in Albemarle's ponds, so at any moment those ponds hold **eighteen to twenty-four months of their own output**, sitting in the sun. The residence time is the inventory li-w1 · validated.

And because the company declines to put a ceiling on how long the sun takes, there is no ceiling on this either.

Is that a buffer? The question has no answer until you say what against. If the brine stops arriving, the ponds keep delivering at full rate for the full residence time and then stop. Against an interruption behind them that is eighteen to twenty-four months in hand; against anything in front of them it is nothing at all, because you cannot pull a tonne forward without giving it sunlight it has not had

li-w1 · validated.

But it buys time, not supply. Run a warehouse and a pond system against the same six-month interruption: the warehouse loses nothing, and the ponds lose all six of those months, but lose them **eighteen to twenty-four months after the brine stopped.** The outage still arrives, undiminished. It only arrives late li-b1 · validated.

Postponement is worth a great deal. It is not absorption, and this chapter will not let the one read as the other.

And a warehouse differs from the ponds, but not by running down gently: anything drawn at full rate ends in a cliff, and rationing is a decision somebody makes rather than a property of the stock. The real difference is that **in a warehouse the level is a choice, and in the ponds it is an identity**: residence time times throughput, and nobody gets a vote.

A derivation, not a measurement: a sourced residence time run through a standard identity. It stays a *ratio* permanently rather than for want of a source: the ponds destroy most of their own mass on purpose, so a tonnage would first have to choose between raw brine, contained lithium and finished product, three answers orders of magnitude apart. It also assumes the ponds are running level; a growing operation holds less, a shrinking one more.

And the question a reader most wants, how many days between a break and something changing at a wall socket, turns out to have been the wrong question, in an instructive way. There is no one answer, because there is no one break. Cover is not a property of a chain. It is a property of a cut li-b1 · validated.

Ask it cut by cut and the chain splits cleanly in two. **How long until a break is felt can be computed from physics wherever somebody publishes a transit time. How much of the break gets absorbed is unpublished at every single link.** This chain's delays are public, because they are physics and operators describe their own processes. Its inventories are private, because they are commercial

li-b1 · validated.

So the chapter can say precisely which half of the buffer question this chain lets anyone ask, and the one cut it can answer in full, it answers from the sun rather than from anybody's disclosure li-b2 · validated.

This chapter was **held** until both of those clocks were ruled. They have been, and it is released Editor · 2026-08-25.

A chapter that lists what it cannot say is not being modest. It is the only way the things it *can* say stay worth anything.

Who: four kinds of party, because at this stage the unit changes. The owner, a utility or an independent operator. The public utility commission that approved the installation and decided how its cost is recovered, which is where the cost decision is actually made, and a body whose hearings the public can attend. The ratepayers of the territory it serves, who pay for it. And whoever lives beside the site, named separately, because living next to something is not the same as paying for it. No figure is put on anyone's bill: nothing in this study supports one. Every other 'who' in this study is somebody else: a mine's workforce, a smelter's neighbors, a town along a fence line. At the end of this chain the reader stops being a spectator and becomes a party, because the installation is paid for through the rate base they are in. store-g1 · validated

What you would actually feel

AFTER SIX CHAPTERS · WHAT YOU WOULD ACTUALLY FEEL

Six chains, six chapters, and one question underneath all of them: **if a link went, when would you know?**

This chapter answers it from the record (every case below is something that happened, with dates), and it answers it in a shape that surprised the people who built it. **There is no single answer, because there is no single break.**

It also declines something the study set out to do. We meant to say which interruptions were most likely. We cannot: the cases here are in the study *because they happened and were written down*, which is no basis for a rate, and the hazard data that does exist covers hurricanes and earthquakes far better than it covers a fire at the only crucible works. **An ordering built on it would rank these chains by how well studied their hazards are** dis-p1 · validated.

SIX CHAINS, TWO QUESTIONS

Ask each chain how long a break takes to reach you and the six do not answer the same way. **They do not even answer the same kind of question**, and sorting out why took longer than any single number in this study.

What settled it splits in two. **Physics decides what kind of answer a chain can give about its own disruption; the record decides whether we have it. More study fills the second and never converts one kind into the other** dis-s1 · validated.

Copper: a reservoir feeding a process that can be slowed, so the answer is a curve rather than a number; and we hold the shape, not the volume. **Lithium:** a pipeline transit nobody can hurry; computed from physics, because no interruption of it is on record. **Oil:** days of cover in a drawable reservoir, and the only chain where an agency publishes both halves of that fraction. **Iron:** a queue, because the thing is built to order and nothing substitutes; a lead time, and no failure clock at all. **Salt:** days of cover at the point of use, observed, and the only chain with three endpoints dated. **Sand:** days of cover spread along the chain; observed, and a halt somebody chose.

These are not six sizes of the same thing, and they cannot be put in order. **An ordering, like a ratio, presumes a common origin, and these six do not share one.** A lead time runs from an order placed; a restart from a closure; a residence time from material entering a pond; days of cover from the moment resupply stops. Ordering them would order the definitions, not the chains. **The ordering is not forbidden here so much as meaningless**, and the shared unit is the hazard rather than the help, because days on all six is exactly what makes the sort look available dis-s1 · validated.

ONE STORM, TWO VALLEYS

Late September 2024. Hurricane Helene reaches the southern Appalachians and two of this study's chains lose a node in the same week: a quartz district in one valley, a saline plant in another, roughly fifteen to twenty kilometers apart sand-x1 · validated; salt-x1 · validated.

They came back on completely different clocks. Spruce Pine halted on September 26 and announced a restart about **fourteen days later** spine-f1 · validated. The flooding arrived on the morning of the 27th helene-f1 · validated, and North Cove closed on the 29th, back to its pre-storm output **a hundred**

and forty-one days after that `nc-f1 · validated`.

And the plant coming back is not the shortage ending. That is the distinction this chapter exists to make. The first line restarted at thirty-two days; all ten at a hundred and twenty-one; the output back where it began at a hundred and forty-one, and the shortage itself was still being reported to the Senate about **eleven months** after the water came in `nc-f1 · validated`.

One driver, two chains, not two storms' worth of evidence. The quartz mine and the saline plant share nothing but weather: different owners, different regulators, different customers, and neither one reached the other. And Sibelco never published a date for full capacity, so the two chains' *full* recoveries cannot be compared at all: only their restarts, and only their first lines.

THE CLOCK NOBODY CAN HURRY

Lithium has no interruption on record, so its answer is computed rather than watched, and the physics is unusually willing to be computed. Brine spends **about eighteen to twenty-four months** in the evaporation ponds `li-e1 · validated`, so at any moment the ponds hold that much of their own output, sitting in the sun `li-w1 · validated`.

It looks like an enormous buffer and it does not behave like one. If the brine stopped arriving, the ponds would keep delivering at full rate for the full residence time and then stop. Run that against a six-month interruption and a warehouse loses nothing while the ponds lose all six months of it, **not for another eighteen to twenty-four months, and then in full.** The outage still arrives, undiminished. It only arrives late `li-b1 · validated`.

Postponement is worth a great deal. It is not absorption, and the difference is the whole of what a pipeline is.

THE MINE THAT SLOWS

Copper's answer is the one that refuses to be a number at all. The Escondida concentrator runs on desalinated water pumped up three thousand meters from the coast `esc-c1 · validated`, and if that supply were interrupted, **the mine would not stop. It would slow.**

So there is no single figure to give. Cover becomes a relationship between how much water is left and how hard the plant is being run, and **the slowing is what would reach the market** rather than any stoppage `dis-s1 · validated`.

We hold the shape and not the size: how much water is stored on site is a figure we do not have, so how long the slowing could last, **or when it would become a stop**, is a question this study cannot answer `esc-w1 · validated`.

THE ONE SOMEBODY COUNTS

Oil is the exception that shows what the others are missing. A federal agency publishes the stock *and* the rate it is consumed at, every week, so the buffer can simply be divided out.

US distillate cover in 2026 runs at **about twenty-seven to twenty-nine days**, measured against distillate supplied and averaged over four, thirteen and fifty-two weeks, against about thirty-two to thirty-four days in 1996, measured the same way `oil-b2 · validated`.

The same barrels buy about four to five fewer days than they did in 1996, taken basis by basis rather than by subtracting the printed ranges `oil-b2 · validated`.

Worth noticing what makes this chain legible: not that its buffer is better understood, but that **somebody publishes both halves of the fraction**. No other chain here has that, and none of them is less important for it.

A QUEUE, AND HALF A CLOCK

The largest transformers are built to order, and nothing substitutes for one. So the chain's answer is not a buffer at all: it is a **queue**. The largest transformers are quoted at around **three years, and up to five**, a figure counted from the order being placed, not from a failure, and one the Department of Energy carries from its own conversations with domestic producers rather than from a measured series xfmr-f1 · validated.

And the other half of the pair is missing. A transformer failing is an event with a duration, and this study found no published clock for it, so where every other case here has a stop and a return, iron has a return and a blank.

In the study's paired-bar exhibit that blank is drawn rather than hidden: the left-hand bar runs open, because an endpoint we do not have is drawn as an endpoint we do not have.

There is an escape hatch, and it is a club. A national reserve was studied in 2015 and declined in favor of voluntary sharing among the utilities that own the transformers. The sharing is real, funded and federally cataloged, and of its five programs, exactly one adds a transformer to the country that would not otherwise exist, and the two most formal had never been used as of DOE's 2024 report grid-g1 · validated.

Whether the club is enough is unproven, and not provable from the public record, because the one dedicated inventory is sized, in its operator's own consultant's model, to a 98 percent likelihood of meeting anticipated need, and neither the inventory nor the model's inputs are public. What can be said is who it reaches: members whose regulators let prevention be paid for. The study takes no position on what should exist instead.

THE CASE WHERE THE CLOCK DOES NOT END

Every case above has a clock that stops. A restart announced. An output restored. A shortage declared over. A transformer delivered. Even the pond transit ends.

Mossville does not. An unincorporated community in Calcasieu Parish, founded over two centuries ago by free Black families led by the freedman Jim Moss; industry arriving from the 1940s; and in 2013 a voluntary property-purchase program that bought it out. **There is no back-to-normal for a community that no longer exists as a place** pvc-g1 · validated.

This study spends four chapters on instruments that arrive late and arrive public: a fenceline monitor, a pressure gauge, a published quarter study-g1 · validated. **Every one of them assumes somebody is still there to read it.** A community that has been bought out cannot consult a monitor, cannot re-file a complaint, cannot hold a seat at a table. It cannot be the public that public data is public to.

And there is a second way that assumption fails. Every one of those instruments also assumes it turned up. In this corridor one of them has not: the fence-line monitoring these plants were to begin in July 2026 was deferred for two more years, by a proclamation filed on the morning it took effect, whose Annex I lists the plants exempted chl-g1 · validated. **Mossville is where the reader is gone. Annex I is where the reader is there and the instrument has not arrived.**

The buyout is Sasol's, and Sasol runs a cracker and a gas-to-liquids plant, **not the vinyls chain this chapter follows**. It is named because it is the limit case of the study's own question about what is transient and what redraws a map, not because it is this chain's doing.

WHAT THIS CHAPTER CANNOT SAY

It cannot tell you which break is coming. Not because nobody has looked, but because the cases are here on account of having been documented, and a set assembled that way carries no rate `dis-p1 · validated`.

It cannot tell you what most of these chains are holding. One publishes its stocks weekly. For the rest, the delays are public (physics, and operators describing their own processes); **the inventories are private where they are commercial, and simply uncollected where they are not**

`li-b1 · validated`.

And it will not tell you how a shortage felt to the person at the end of it, because the record does not contain that either. What happened to patients during the saline shortage is recorded in a survey of their pharmacists `bed-g1 · validated`. A community can be bought out and move; a workforce can quit; a firm can hold its feedstock somewhere else. **A person in a bed connected to a line can do none of those, and had no voice in the record either.**

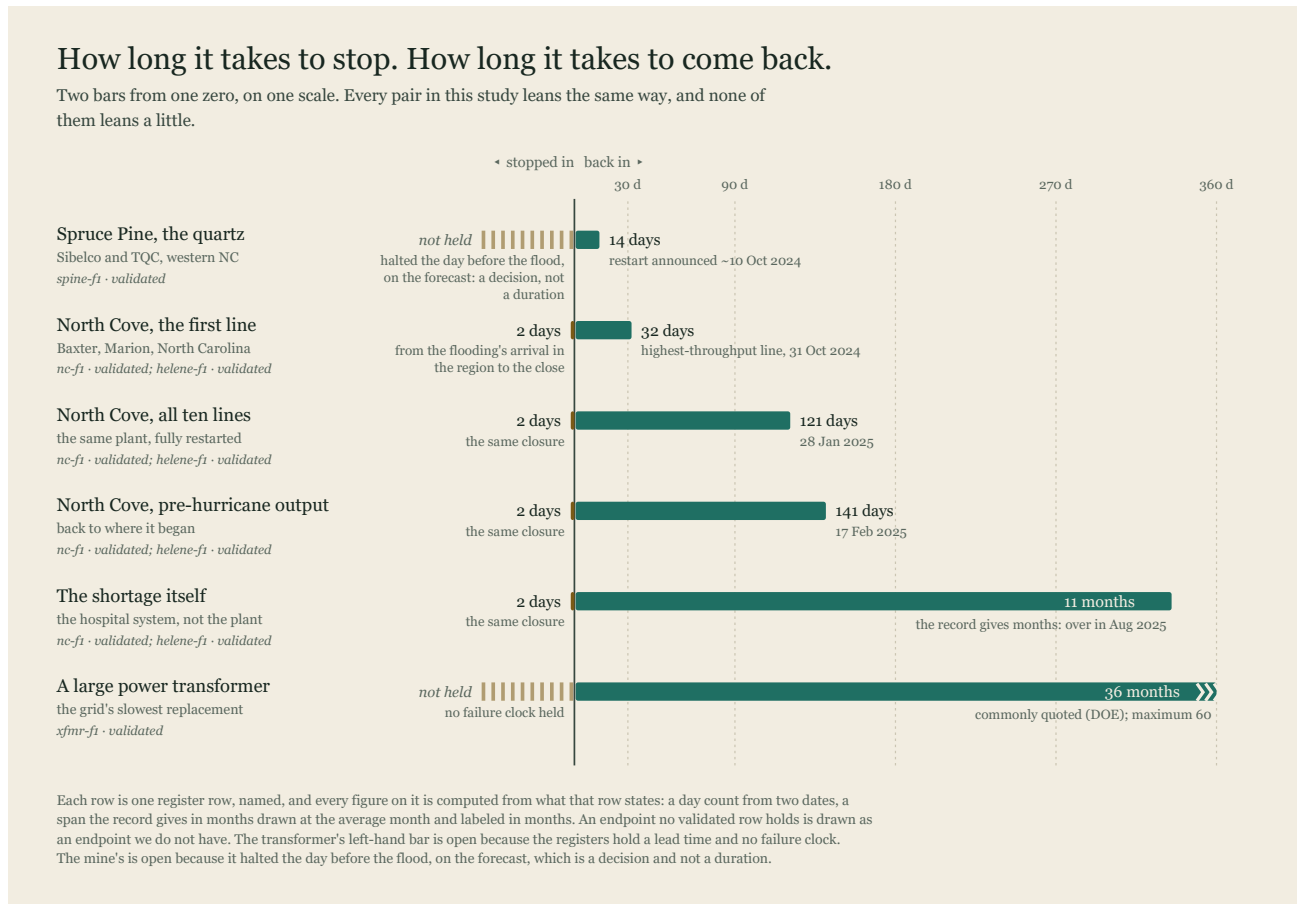
Six chapters were built to make one question answerable, and the honest answer is a shape rather than a number: how long a break takes to reach you depends on where the stock sits and whether anyone can hurry it, and on whether anybody thought to write it down.

What it adds up to

THE SHAPE EVERY CHAPTER MADE

Three chains and six pairs, and most of them are one week. A hurricane emptied a stretch of western North Carolina: a quartz mine halted ahead of it, a saline plant closed as the rain arrived, and **four of these rows are the same plant**, asked four different ways about when it was back. The sixth is not an event at all. It is a queue.

The silhouette is the same every time. A thing stops in a day or two. It comes back in a season, or in years.



Stop against return, on one scale. The transformer's bar leaves the page rather than the page being rescaled to hold it `xfmr-fi · validated`, and its left-hand bar is open because the registers hold a lead time and no failure clock.

That asymmetry is not a fact about storms. It is what a buffer is for, and what it costs to have spent one: stopping is a decision anyone can make in an afternoon, and restarting is a queue `nc-fi · validated`.

The oil chapter arrived at the same shape from the other end, not a pair of bars but a buffer measured in days of cover, and the same barrels now buy about four to five fewer days than they did in 1996 `oil-b2 · validated`. Lithium still cannot join it, and the reason has changed: its two clocks are now ruled `li-b1 · validated` `li-b2 · validated`, but they are STRUCTURAL, what the process must take, and every row above is a realized event with two dated ends. There is no lithium event in these registers to draw.

THE RECEIPT

The building wire in one average American house, which is what every row underneath this study actually measures. Here is what it took, in the only units this study trusts.

The price is not the total of the lines above it. It is the other ledger, money rather than quantities, set apart below them: the metal in it, about \$840 at the 2024 average US producer cathode price, and not the wire referee ruling.

The people are named as roles, never as a count. Six of this study’s Who rows hold six different units (a site, an operation, a company, a legal bargaining unit, a forecast, an exposed population), and no total can honestly be formed from them seat ruling. The seafarers on both crossings are not counted in this study, so the receipt says that too.

A receipt makes everything on it look settled. That is exactly why this one prints what it cannot say, and why the study will not put a number on a line it has not earned: the format would launder it.

THE BUILDING WIRE IN ONE HOME	
one average US single-family home · what it took	
copper in the wire	88 kg
195 lb of building wire; the source leaves the figure undated	dfw-f1 · validated
energy to make it	0.91-1.15 household-months
counted in months of a home's electricity use; 0.91-1.15 on the ruling's mixed basis, 0.35-1.15 across bases	dfw-c1 · validated
the long leg	about 6,700 miles
(10,760 km) Shanghai to Los Angeles, port to port	col-d1 · validated
fuel for the sea legs	3.6-5.3 gallons
bunker, this home's wire, both crossings; sea legs only	rh-bunker-comp · validated
if it is ever recycled	not measured in this study
house wire comes out as insulated scrap and goes back through a smelter; only clean scrap at the mill is measured here, and this is not it	see swr-c1
at sea, both crossings	48-52 days
at each Fleet's average speed, the ore's leg and the metal's; figures that lean long	route-t1 · validated; route-t3 · validated
elapsed, ore to wall	not established
time in port, at the smelter, overland and at the mill is not sourced in this study	
hands it passed through	
miners and contractors · smelter crews · seafarers · overland freight crews · the wholesale tier · licensed electricians · homebuilders	
roles, not a headcount: the rows hold six incommensurable units and no total may be formed from them	
rests on: esc-f1 · validated; gux-w1 · validated; sea-w1 · validated; move-w1 · validated; dfw-w1 · validated	
not counted in this study: the seafarers on both crossings	
borne by	
the Peine community, Atacama · farming villages in the Guixi deposition plume	
borne by, not passed through: people outside the chain who did not consent to it and cannot leave it	
esc-g1 · validated; gux-f1 · validated	
AND ON THE OTHER LEDGER	
what the metal in it was worth	about \$840
the copper in it at the 2024 average US producer cathode price; the metal, not the wire, which carries drawing, insulation, distribution and margin this figure does not. The quantity it multiplies is undated at its source	dfw-p1 · validated
Every figure names the register row it rests on. The price is not a total of the lines above it: it is the other ledger, money rather than quantities, set apart below them. The lines marked not measured, not established and not counted are what it does not know, and says so.	

The receipt, the static edition: what the building wire in one average US home took, who handled it, who bears it, what the metal in it was worth, and the lines it does not know. The tour prints this line by line; the artifact is the same either way.

THE WHOLE BOARD

That's one wire, one machine, one chip and one bag. The study runs this chain for six materials (a chip, a saline bag, this transformer, that wire, a tank of diesel, a battery), then asks the questions the map raises: how deep are the buffers, how fast is a break felt, and which breaks redraw the map for good. Explore

freely: hover a node for its name, and every dotted node opens.

Appendix A · the register ledger

Every row the study keeps. Standing runs open, sourced, validated, integrated. For a validated row the last column says what was validated, in words its validating seat has read. For a row below validated it gives the question the row was opened to ask, labeled as a question; such a row carries no claim in the published chapters. Generated from the registers at build time.

copper · 83 rows

ROW	STANDING	VALIDATOR	WHAT IT HOLDS
den-world-mine	validated	-	
den-world-refined	validated	-	
den-lifecycle-energy	validated	-	
esc-f1	validated	institutions-commons	Minera Escondida's own statutory annual report: 4,000 employees of its own and 6,532 operational contractors at December 31, 2025. Two figures, never summed. The filing does not say whom 'operational' leaves out.
esc-f2	open	energy-biophysical-economics	<i>question asked:</i> truck-fleet diesel consumption (or fleet size) from BHP/primary
esc-f3	validated	author	
esc-c1	validated	-	Desalting the mine's water takes about 3.4 kWh per cubic meter and pumping it 3,100 meters up and 170 km (about 105 miles) inland about 14 (band 11 to 15), so the climb costs about four times the desalting; at the plant's stated flow that is about 1.4 TWh of electricity a year.
esc-c2	validated	econ-stats-referee	Escondida produced about 5.7% of the world's mined copper in its 2025 fiscal year, one tonne in eighteen (company filing over the USGS world estimate); the share is of mined copper, not refined, and the year before it was nearer one in twenty.
esc-g1	validated	institutions-commons	The Peine conciliation, from the tribunal's own record: the community filed before the state did; the agreement was approved in December 2024 on a divided bench, the dissent holding that no measure repairs the drawdown itself and that recovery around the year 2200 assumes a cessation nobody signed; about US\$47 million and fourteen measures, among them a standing table where the community sits; implementation uneven as of early 2026. Characterized by venue and status; never as resolved.
esc-f4	validated	econ-stats-referee	
esc-f5	validated	energy-biophysical-economics	Chile's national electricity use per tonne of copper in concentrate, 2024, with its 2010 baseline, and open-pit fuel use per tonne (Cochilco). National averages by process; not any one mine's.
esc-f6	dropped	econ-stats-referee	<i>dropped; carries no claim</i>
esc-f7	sourced	econ-stats-referee	<i>question asked:</i> desal investment (audit A8; distinct from col-f2 capacity)
esc-f8	open	energy-biophysical-economics	<i>question asked:</i> Escondida/Spence PPA split, if BHP filings disclose (energy memo rec 5 — collapses apportionment estimate)
esc-w1	validated	complex-systems-polycrisis	A water reservoir exists at the Escondida mine, between the pipeline from the coast and the concentrator; the operator's 2013 filing names it and gives no capacity. Its volume, and so any number of days of cover, is not established and never renders as a quantity.
esc-g2	open	institutions-commons	<i>question asked:</i> NEW (instcommons rec 7): pull the approval resolution PDF (fojas 32157) — full Mo-M13 obligations, timelines, breach machinery; fills 'graduated sanctions: not ...

esc-ghost1	sourced	econ-stats-referee	<i>question asked:</i> Same-scale ghost: Manhattan Island outline over the Escondida district (concept F, ghost #1 of the two-ghost budget)
esc-f9	open	econ-stats-referee	<i>question asked:</i> Copper ore-grade decline series: ~40% (19th-c Montana) -> ~4% (early-20th-c US avg) -> ~0.4% (now) — spoken by Hagens, unchallenged by Conway (TGS ep.127) ...
esc-f10	open	econ-stats-referee	<i>question asked:</i> Conway (TGS ep.127): net zero requires 'another three [Chuquicamata-scale] mines every year' to 2050 [SECONDARY-INTERVIEW -> verify vs IEA critical-minerals reports ...
esc-i1	validated	-	
esc-i2	validated	-	
col-f1	validated	energy-biophysical-economics	The mine's concentrate leaves by slurry pipeline to its own port (operator presentation, 2004); this seat validated one thing only, that the concentrate runs downhill from about 3,100 meters while the desalinated water is pumped up the same corridor, and nothing about whether the line is pumped.
col-f2	sourced	energy-biophysical-economics	<i>question asked:</i> EWS desal plant capacity from BHP/Bechtel primary
col-i1	integrated	author	
col-d1	validated	energy-biophysical-economics	Port-to-port distances for the three ocean legs (NGA Pub. 151); the longest, Shanghai to Los Angeles, is 5,810 nautical miles: 10,760 km, or about 6,700 miles.
route-d8	validated	energy-biophysical-economics	The concentrate's leg from the port to the smelter, and the cathode's back, are each at least about 280 miles (450 km) in a straight line; the routes themselves are longer.
route-d7	validated	energy-biophysical-economics	The finished wire's last leg is at least about 650 miles (1,050 km) in a straight line, from the mill at Carrollton to the Dallas area; the route itself is longer.
route-d6	validated	energy-biophysical-economics	The cathode's US leg is at least about 1,850 miles (3,000 km) in a straight line, from Long Beach to the mill at Carrollton; the route itself is longer.
col-g1	validated	institutions-commons	Caleta Coloso's fishers' organizations administer the cove under Chile's Caletas law from November 2025 (the fisheries agency's own account), and the port's owner funds the cove's new facilities through its foundation (the company's account, with the fishers' president quoted on the new infrastructure on the agency's page). Who will own the facilities, the agreement's term, and any exchange behind the funding are not established.
route-t1	validated	energy-biophysical-economics	At the container fleet's average speed at sea, the 5,810 nautical miles (about 6,700 miles) from Shanghai to Los Angeles is 14 to 15 days; a fleet-wide annual average, not a voyage record, and it leans long because a scheduled ocean crossing tends to beat its fleet's mean.
route-t3	validated	energy-biophysical-economics	At the bulk-carrier fleet's average speed at sea, the ore's 9,150 to 9,700 nautical miles (about 10,500 to 11,200 miles) from northern Chile to Shanghai is 33 to 37 days, more than twice the metal's crossing, because the ore travels farther on a slower kind of ship; the same fleet-average caution applies.
route-t2	validated	energy-biophysical-economics	
gux-f1	validated	institutions-commons	A peer-reviewed study documents arsenic, cadmium, copper, lead and zinc in the soils and vegetables of named villages around the Guixi smelter. The scale, about fifteen villages and ten thousand people, is a remediation organization's figure. This study states no health outcome.
gux-f2	validated	econ-stats-referee	Jiangxi Copper's own 2024 annual report lists its provincial state parent as the largest shareholder at 44.06% at year end; that is one holder in a top-ten table, and no sum across holders is used for the owner-operator point.

gux-c1	validated	-	Smelting copper concentrate to cathode took about 9 GJ per tonne as a 2017 China average (modern band 9 to 14), against 250 to 400 kWh per tonne for the electrolytic refining step alone; national averages, not the Guixi plant's own figures.
gux-c2	validated	econ-stats-referee	Guixi is the only single plant that refines more than a million tonnes of copper a year, by international capacity tables and the operator's own annual report; the claim is about refining, and the plant is not the world's largest smelter.
gux-f3	dropped	UNCONVENED-EARTH-SYSTEMS	<i>dropped; carries no claim</i>
gux-f4	validated	econ-stats-referee	
gux-i1	validated	-	
cath-i1	integrated	author	
swr-f1	validated	econ-stats-referee	Southwire describes itself, on its own website, as one of the leading manufacturers of wire and cable; that is the company's word and its own account, and no rank, share, headcount or revenue is established.
swr-f5	validated	author	That Southwire operates a Building Wire Plant in Carrollton, Georgia, beside its Copper Rod Plant (the operator's own newsroom, December 2025). A plant and its purpose; no process step, no tonnage.
swr-w1	validated	institutions-commons	Rod-mill and building-wire-plant crews work at Southwire's Carrollton works: a role at plants the operator itself names, and no more. No count, and nothing about where they live.
swr-f2	validated	energy-biophysical-economics	What the maker specifies for the shaft furnaces it sells: fuel 'natural gas and/or LPG', and 'up to 20% high quality scrap' in the charge (Southwire product sheet and SCR systems page). A product specification; not a statement about the Carrollton furnace.
swr-c1	validated	energy-biophysical-economics	Clean copper scrap charged to a shaft furnace reaches molten copper for roughly three to six percent of the ore route's primary energy. Excludes collecting, sorting and shipping the scrap; applies to clean scrap only; one term in it is assumed.
swr-f4	open	energy-biophysical-economics	<i>question asked:</i> Energy intensity per tonne of the Carrollton rod line (melt + roll + draw). Split from swr-f2.
swr-i1	integrated	author	
swr-f3	validated	econ-stats-referee	Southwire states that its own Carrollton rod mill is rated at 54 tonnes an hour; that is a rated figure in the company's words, the page does not say which of its Carrollton mills it describes, and no annual output is established.
dfw-f1	validated	-	
dfw-b1	validated	econ-stats-referee	What a wire distributor keeps on hand is not published in the federal monthly wholesale survey, which reports only the parent category, electrical and electronic goods, and not the wiring-supplies line (the annual survey was not checked); and even that line would be dollars of stock over dollars of sales, which moves with the price of copper and is not a count of wire, so no stock of wire and no days of cover are established here.
dfw-p1	validated	econ-stats-referee	The copper in one average American home's building wire, at the year's average US producer cathode price, was worth about \$840 in 2024; that is the metal, not the wire, which carries drawing, insulation, distribution and margin this figure does not.
dfw-u1	sourced	econ-stats-referee	<i>question asked:</i> where American copper actually goes: the end-use split, and the reason copper's break is felt as a building that waits rather than a light that goes out
gux-w1	validated	institutions-commons	Smelter and refinery crews work at the Guixi works: a role at a named operator's named plant, and no more. No count.

sea-w1	validated	institutions-commons	Seafarers crewed both ocean crossings: entailed by the two sea legs. Role only. This study holds no count of them, and the receipt says so on its own line.
move-w1	validated	institutions-commons	Overland freight crews carried the wire's copper from port to mill and from mill to home: entailed by two inland endpoints. The mode of the port-to-mill leg is not sourced. Role only, no count.
dfw-w1	validated	institutions-commons	Three kinds of hands exist at the install stage, as classes and no more: the wholesale tier, licensed electricians and homebuilders. The household is not a hand. No count and no place.
dfw-c1	validated	-	Making the 88 kg of copper wire in one house took energy equal to about one month of that home's electricity use (0.35 to 1.15 months depending on whether primary or delivered energy is compared), and over forty years the wire carries several hundred times that; the making is energy of all kinds, not electricity alone.
scrap-f1	open	author	<i>question asked:</i> name at least one facility to earn a site pin, else stays a ring
scrap-f2	validated	complex-systems-polycrisis	
scrap-c1	dropped	-	<i>dropped; carries no claim</i>
scrap-f4	open	econ-stats-referee	<i>question asked:</i> NEW: volume flat but VALUE doubled — \$2.76B (2017) -> \$5.38B (2024), unit value \$2,744 -> \$5,625/t (+105%). Decompose price vs composition (spec-gates pushing sorting ...
scrap-f3	validated	econ-stats-referee	
scrap-g1	validated	complex-systems-polycrisis	US copper-scrap exports to all destinations stayed flat, between about 0.78 and 1.01 million tonnes every year from 2017 to 2024, while their destinations reorganized: China's share fell from about 68% to 10% and came back to 41%, and Malaysia's rose from under 1% to 25% and fell to 7%. Two rules from opposite sides did it, China's 2020 reclassification of high-grade scrap and Malaysia's 2022 purity standard; the 2021 dip is not attributed to Malaysia. This is US exports, not world scrap trade.
rh-esc-power	open	-	
rh-esc-power-p	validated	energy-biophysical-economics	Escondida and the Spence mine together contracted 6 TWh a year of renewable electricity from Enel Generación Chile (from August 2021) and Colbún (from January 2022); the volumes are contractual and annual, shared between two mines, and say nothing about hour-by-hour supply.
rh-esc-power-env	open	energy-biophysical-economics	<i>question asked:</i> ENV/ECO: footprint of those named plants (land, siting, lifecycle) — 'renewable' still stands somewhere
rh-esc-power-econ	open	econ-stats-referee	<i>question asked:</i> ECON: PPA scale, seller, what the contract means for Chile's grid buildout
rh-esc-power-comp	validated	-	
rh-gux-power	open	-	
rh-gux-power-p	validated	energy-biophysical-economics	What powers the Guixi smelter is not on public record: a 1,880 MW four-unit coal station stands beside it (Global Energy Monitor), and whether any of its output reaches the smelter is stated nowhere a reader can check.
rh-gux-power-env	open	energy-biophysical-economics	<i>question asked:</i> ENV/ECO: the power plant's own emissions/water on top of the smelter's documented plume
rh-gux-power-econ	open	econ-stats-referee	<i>question asked:</i> ECON: plant ownership, employment, smelter-town dependency (company town economics)
rh-gux-power-comp	sourced	energy-biophysical-economics	<i>question asked:</i> COMPARISON: one feel-it line for the smelter's draw
rh-swr-power	open	-	

rh-swr-power-p	open	energy-biophysical-economics	question asked: POWERS: Georgia grid mix at the rod mill (Southern Co.) + SCR furnace fuel — what actually melts the cathode?
rh-swr-power-env	open	energy-biophysical-economics	question asked: ENV/ECO: that grid mix's footprint (incl. Vogtle share if material) at the mill's meter
rh-swr-power-econ	open	econ-stats-referee	question asked: ECON: Southwire as Carroll County employer; energy cost share of rod production
rh-swr-power-comp	sourced	energy-biophysical-economics	question asked: COMPARISON: one feel-it line for the rod mill's draw
rh-bunker	open	-	
rh-bunker-p	validated	energy-biophysical-economics	
rh-bunker-env	validated	energy-biophysical-economics	
rh-bunker-econ	open	econ-stats-referee	question asked: ECON: freight cost share of delivered wire price (how cheap moving a tonne across the Pacific actually is)
rh-bunker-comp	validated	-	Carrying one home's worth of copper across the Pacific twice, as ore and as metal, takes four or five gallons of ship fuel (3.6 to 5.3; 5.3 to 7.6 GJ per tonne of copper); sea legs only, measured at the ship, with the overland legs excluded because their mode is not sourced here and their distances are straight-line floors only, from which no fuel follows.

disruption · 6 rows

ROW	STANDING	VALIDATOR	WHAT IT HOLDS
dis-p1	validated	econ-stats-referee	The study does not rank disruptions by likelihood, by ruling and with the Editor's acceptance: its cases are here because they happened and were documented, which gives no rate, so it describes recurrence, exposure and concentration and orders nothing.
dis-p2	validated	complex-systems-polycrisis	
dis-p3	validated	institutions-commons	
dis-p4	validated	energy-biophysical-economics	
dis-s1	validated	complex-systems-polycrisis	The six chains answer the disruption question in different kinds (a curve, a transit time, days of cover, a queue), fixed by where the stock sits and whether the process can be hurried. More data fills in a kind and never turns one kind into another, so an ordering across the six would order definitions. A finding about these six chains, which were chosen and are not a sample.
dis-s2	validated	complex-systems-polycrisis	

iron · 35 rows

ROW	STANDING	VALIDATOR	WHAT IT HOLDS
goes-f1	validated	author	Who makes grain-oriented electrical steel in the United States: Cleveland-Cliffs is the sole active domestic supplier and can meet 12 to 20 percent of domestic demand (DOE 2024), closed at company level; the named works is scoped, not claimed.

but-w1	validated	institutions-commons	That mill crews work at the Butler works, as a role. No count, and no finer role than the operator's source supports.
goes-f2	validated	energy-biophysical-economics	Transformer core steel is rolled to nine thousandths of an inch, which is a catalog grade (0.23 mm, the maker's M-3), and it is that thin because the eddy-current part of a core's loss rises with the square of the sheet's thickness; the cost shares and import shares on this row were not this seat's to validate.
goes-i1	integrated	author	
xfmr-f1	validated	econ-stats-referee	The Department of Energy reports that large power transformers, once delivered inside a year, are now commonly quoted at 36 months with a recorded maximum of 60; these are figures the department carries from conversations with producers, not a measured series, and they give no lower bound.
xfmr-f2b	validated	econ-stats-referee	The Department of Energy describes large power transformers as highly customized and usually custom-built to their use; that language is what is validated, and no count of designs or unit cost is established.
xfmr-f2a	open	econ-stats-referee	<i>question asked:</i> LPT unit cost (the \$14M figure) - split from xfmr-f2, and econ would not spend a round on it
xfmr-f3	validated	econ-stats-referee	In 2019, on the Commerce Department's survey as the Department of Energy reports it, 137 of the 754 large power transformers bought for American use were made domestically, about 18%; the class is units of 100 MVA and above, and four in five were imported complete.
xfmr-f4	validated	econ-stats-referee	American demand for large transformers rose: the Department of Energy puts it at about 750 units in 2019, on a basis of units above 60 MVA, and projects about 900 a year by 2027, while reporting that manufacturers described far larger jumps in their own orders; the report does not reconcile the two, and no growth multiple is established.
xfmr-f5	validated	econ-stats-referee	The average installed large power transformer was about 38 to 40 years old, and about 70% were 25 years or older, on industry estimates the Department of Energy published in 2014; the department's 2022 estimate of the installed fleet is 4,900 to 6,799 units; forty years is a typical design life, not a failure age, and the survival figures in the chapter are the study's own fitted estimates.
xfmr-f6	validated	complex-systems-polycrisis	A 2020 federal survey of 87 domestic transformer-component makers, reported by the Department of Energy, found 89% had difficulty finding qualified or experienced workers and named welding and coil winding among the training needs; a 2024 federal advisory report puts the time from hire to fully trained and productive at six to eight months. No source gives a duration for winding a coil, and none renders.
xfmr-c1	validated	complex-systems-polycrisis	The queue model's input bands (fleet age, failure hazard, demand growth, capacity and build time) were delivered by the energy seat, are loaded and checked when the engine starts, and were reproduced by the Author and the referee. This seat validates that the engine uses them as given. It does not validate the hazard and demand physics behind them, which remain the energy seat's.
xfmr-x1	validated	complex-systems-polycrisis	One manufacturer's environmental declaration puts about 40 tonnes of copper in a 500 MVA transformer, and a second maker's 40 MVA unit carries about six tonnes, so smaller units hold more copper per unit of rating. Two declared design points and a direction, not a band for the class and not a coefficient for the fleet; the copper-to-transformer edge stays parked.
xfmr-f7	validated	energy-biophysical-economics	The grid regulator's published forecasts run from about 2 to 2.5 percent a year nationally to 5.6 percent in the fastest-growing region, and all of them clear the computed boundary; the queue becomes permanent only at sustained growth between roughly $7\frac{3}{4}$ and $10\frac{1}{2}$ percent a year, depending on what counts as cleared, on an optimistic bound and with the forecasts themselves rising from one edition to the next.
xfmr-f8	parked	energy-biophysical-economics	<i>question asked:</i> EIA-860 vintage pull — replaces the Gamma age fit wholesale (1960s-70s installation waves a smooth gamma can't represent); upgrade, not gate
xfmr-f9	validated	energy-biophysical-economics	Hitachi Energy's transformer works at South Boston, Virginia, operating since 1968, announced expansions in October 2022 and September 2025 and broke ground in June 2026; the record gives dates and dollars and no capacity figure, so none is drawn from it.
xfmr-f10	open	energy-biophysical-economics	<i>question asked:</i> Distribution-class intrinsic build time B — own primary needed before the queued second-pass run (both DOE primaries are LPT-only; 1-3 mo placeholder ASSUMED)

xfmr-f11	validated	energy-biophysical-economics	
xfmr-f12	validated	complex-systems-polycrisis	
e3-c1	validated	econ-stats-referee	The queue model's ceiling: even at the fastest order growth any maker has reported, the supply paths on record yield a standard quote of about 24 to 25 months (21 to 28 across the model's uncertainty), reproduced on two independent implementations; it is a ceiling at the most generous corner, never an estimate of any wait, and reaching 36 months would take intake of roughly three and a half to four times the 2019 level.
e3-i1	validated	complex-systems-polycrisis	In a priority-first queue, the priority class waits about the build time at every priority share run, and the standard class holds the queue. A mechanism, never a level.
sig-f1	validated	econ-stats-referee	Orders from buyers who could be jumping the queue are between about one in five and one in twenty (a fitted estimate with a wide envelope), and even at the top of that range the effect on the median quoted wait is at most half a month, so it does not explain the gap between the model's ceiling and the quoted 36.
xfmr-f13	validated	econ-stats-referee	In March 2025 GE Vernova's chief executive said in an interview that the company's order backlog for gas turbines, power transformers and switchgear together stretched into 2028; that one man said this is all that is established, it is not a transformer-line horizon, and no unit order book has been found for any maker.
iq-f1	validated	econ-stats-referee	Power plants and batteries holding signed or draft interconnection agreements came to about seven and a half times what the country connected in 2024 (Lawrence Berkeley's queue data over EIA's additions); these are agreements, not transformer orders, about a third of past agreements were later withdrawn, and the figure is context for the queue, not a forecast.
xfmr-f14	open	energy-biophysical-economics	<i>question asked:</i> US domestic LPT production path 2020-24 — the un-primaried half of supply after the import side went realized-history
xfmr-f15	validated	econ-stats-referee	Transformer prices rose about 80% from the start of the pandemic on a quoted-price basis (a federal advisory council, single source) and 63 to 68% on the federal producer price index for the industry; the two bases measure different things and both are given.
xfmr-f5b	open	econ-stats-referee	<i>question asked:</i> DISTRIBUTION-class ageing: ~75% of distribution transformers past expected service life - split out of xfmr-f5, which asserted it on LPT evidence
age-i1	integrated	author	
move-f1	validated	author	What moving a large power transformer involves: units of roughly 100 to 400 tons, and only about thirty Schnabel railcars in North America (DOE 2014 and 2024, verbatim).
grid-g1	validated	institutions-commons	A federal report in 2017 weighed a strategic transformer reserve and recommended industry-led sharing instead, and DOE's 2024 report did not change that. Of five cataloged sharing programs one adds a transformer to the country, and the two most formal had never been used as of 2024. Utilities reported holding spares equal to 116% of the most-crucial units in 2016. DOE names rate recovery as a barrier to joining the one program that adds stock. That program's operator holds its inventory to a 98 percent likelihood of meeting anticipated need in its own consultant's model (DOE 2024, p. 31); the inventory is not disclosed and DOE's report does not give the model's inputs, so the figure cannot be checked from the public record. The pooling curves are a model's and are good for direction and order only; that a club has its weakest incentive to admit those who would gain most is an inference, not a computed result. The study takes no position on a federal reserve.
grid-g2	validated	complex-systems-polycrisis	
grid-f1	validated	author	
grid-f2	validated	complex-systems-polycrisis	A federal document, citing media reports, says the 2013 incident at a California transmission substation was met by a re-dispatch that avoided a blackout and that the damage took nearly a month to repair. That attributed restatement is the ceiling: no primary source exists, and neither a count of transformers nor a figure in days is established.

sst-f1	open	energy-biophysical-economics	<i>question asked:</i> SST entrants (Amperesand, Heron Power, DG Matrix), supply-chain contrast (SiC/ferrite vs GOES), Nvidia 800-VDC architecture pull [SECONDARY-NYT -> company/Nvidia primary]
sst-f2	open	energy-biophysical-economics	<i>question asked:</i> data-center load whipsaw (training-mode 500 MW swings) as grid-stability driver [SECONDARY-NYT -> grid-operator primary]

lithium • 20 rows

ROW	STANDING	VALIDATOR	WHAT IT HOLDS
li-x1	validated	complex-systems-polycrisis	One community and one tribunal treated a copper operation and a lithium operation as jointly responsible for one named aquifer, and the December 2024 agreement binds them to shared measures and a standing table. The edge between the two chains is institutional. Whether pumping brine and pumping fresh water act on the same water is disputed and is not settled here; a disruption to copper does nothing to lithium.
li-e1	validated	energy-biophysical-economics	Albemarle says of its own ponds on the Salar de Atacama that evaporation takes about eighteen to twenty-four months and may be prolonged by weather; one operator's figure for one operator's ponds, and the other operator on the same salt flat publishes none.
li-w1	validated	econ-stats-referee	Because brine spends eighteen to twenty-four months in Albemarle's evaporation ponds, by the company's own filing, those ponds hold about that many months of their own output at any moment; this is a derivation from that residence time, stated as a ratio and never a tonnage, it assumes steady operation, and the company sets no upper limit on the time.
route-d3	validated	energy-biophysical-economics	The lithium chain's two sea distances, reused from the copper chain's and carrying its conditions: 9,150 to 9,700 nautical miles (16,946 to 17,964 km, or about 10,500 to 11,200 miles) from northern Chile to China for the carbonate, and 5,810 (10,760 km, or about 6,700 miles) from Shanghai to Los Angeles for the packs (NGA Pub. 151); nothing new is measured.
route-t5	validated	energy-biophysical-economics	At the container fleet's average speed at sea, the packs' 5,810 nautical miles from Shanghai to Los Angeles is 14 to 15 days, a figure that leans long; that this trade crosses by sea in containers is on the federal record (US Census, 2024: 99 percent of HS 8507 from China by weight went by vessel, and 99 percent of that containerized), which establishes how the trade moves, not how one product does.
route-t6	validated	energy-biophysical-economics	At the container fleet's average speed at sea, the carbonate's 9,150 to 9,700 nautical miles from northern Chile to China is about three to three and a half weeks (22 to 25 days), a figure that leans long; the containerized hull is what SQM, the salt flat's other operator, describes for its own product in its annual report, which also names a slower break-bulk alternative that is not quantified here.
li-c1	validated	energy-biophysical-economics	Mine production of lithium in 2024: Australia 36.7 percent, Chile 20.4, China 17.1, the top three together 74.2 (USGS, lithium content); these are shares of mining, not of refining, and of a world total from which the United States' output is withheld.
li-c2	validated	econ-stats-referee	Read off one published International Energy Agency chart for 2024, to within a point or two: China refines about 70% of the world's lithium and the top three countries about 96%, against about 44% and just under 60% for copper; these are readings of a chart, not measurements, battery-grade graphite and magnet rare earths sit higher still, and no comparison with the study's other four materials is established.
mine-g1	validated	institutions-commons	Who: nobody, and the reason is worth a sentence. This stage is a world total, not a place, and adding up is the operation that removes people. Five producing countries stand behind the figure, with five workforces, five sets of neighbors and five regulators; the number sums their tonnes and nothing else. Where the chain reaches a place again, the people return.
li-n1	sourced	econ-stats-referee	<i>question asked:</i> The carbonate plant this chain's brine goes to is named in the record, and it is Albemarle's, at La Negra
lng-g1	validated	institutions-commons	Who: Albemarle runs the carbonate plant, and it is also one of the companies in the Peine settlement at the salt flat. The company links the two places; the community does not. Peine's case is about groundwater at the salt flat, not about this plant. Around the plant is an industrial district of the Antofagasta region with people living in it, and this study can say no more about them than that: it holds no Chilean regulatory record for the plant itself.
li-p2	dropped	complex-systems-	<i>dropped; carries no claim</i>

polycrisis			
cam-g1	validated	institutions-commons	Who: the makers of cathode powder are specialty chemical producers. This study names none of them and has not sourced where their plants are. The work is preparing the precursor, firing it in kilns, and quality control. No headcount is given, and this study holds nothing on the communities beside these plants. A class, standing in until a plant is named.
cell-g1	validated	institutions-commons	Who: the makers of lithium-ion cells. This study names none of them and has not sourced where their plants are. The work is coating electrodes, assembling cells, and the slow first charging and testing that follows. No headcount is given, and this study holds nothing on the communities beside these plants. A class, standing in until a plant is named.
li-p1	validated	author	Where one named product's modules and packs are assembled, from the maker's own annual filing (Lathrop, California and Shanghai). One product; not a statement about the industry.
pack-g1	validated	institutions-commons	Who: the firms that build cells into modules, packs and containers. Where that is done is not established in general. In the one case this study has sourced, the maker's own annual filing names dedicated factories for the product in Lathrop, California and in Shanghai: neither where the cells are made nor where the battery is installed. Other makers may do it differently. The work is module assembly, enclosures, cooling and commissioning. No headcount is given. This study has not looked for a community record at either factory, so it states none.
li-b1	validated	energy-biophysical-economics	How much warning a break gives can be worked out from how long each process step takes, wherever that time is published; how much stock stands ready to absorb a break is published nowhere in this chain, in tonnes or in days.
conv-11	validated	energy-biophysical-economics	
li-b2	validated	complex-systems-polycrisis	Restarting a chain behind an evaporation pond cannot be faster than the pond's residence time, whatever anyone does: the pipeline that delays a break by its residence time charges the same time again on the way back. A floor given by physics, on the residence time the lithium chapter sources; no interruption is on record to observe it.
store-g1	validated	institutions-commons	Who: four kinds of party, because at this stage the unit changes. The owner, a utility or an independent operator. The public utility commission that approved the installation and decided how its cost is recovered, which is where the cost decision is actually made, and a body whose hearings the public can attend. The ratepayers of the territory it serves, who pay for it. And whoever lives beside the site, named separately, because living next to something is not the same as paying for it. No figure is put on anyone's bill: nothing in this study supports one. Every other 'who' in this study is somebody else: a mine's workforce, a smelter's neighbors, a town along a fence line. At the end of this chain the reader stops being a spectator and becomes a party, because the installation is paid for through the rate base they are in.

oil · 25 rows

ROW	STANDING	VALIDATOR	WHAT IT HOLDS
oil-b1	validated	energy-biophysical-economics	
oil-b2	validated	energy-biophysical-economics	US distillate stocks divided by distillate product supplied gave about 27 to 29 days of cover in mid-August 2026 against about 32 to 34 in the same week of 1996, measured the same way over four-, thirteen- and fifty-two-week averages; a ceiling on comfort, not a countdown, since tank bottoms and pipeline fill are counted in the stock.
conv-o1	validated	energy-biophysical-economics	
route-o1	validated	energy-biophysical-economics	
route-d4	validated	energy-biophysical-economics	The diesel's last leg is at least about 250 miles (400 km) in a straight line, from Port Arthur, where the map places the refining, to the Dallas area; the route itself is longer.

oil-s1	validated	econ-stats-referee	The Roundtable's first study found that a severed diesel link adds a median of about \$23 billion a month to American household costs, falling at 8.9% of after-tax income at the bottom of the income distribution against 1.6% at the top; it priced a shock and modeled no delay to any delivery.
oil-g1	validated	econ-stats-referee	
who-up1	open	institutions-commons	<i>question asked:</i> Name the operator CLASS and the post-2020 consolidation - the ownership-side view of oil-ci's U
who-up2	validated	institutions-commons	The Railroad Commission of Texas, by its own account, was created in 1891 to regulate railroad charges and passed its last rail duties to the state transport department in 2005. New Mexico's Land Grant Permanent Fund dates from statehood in 1912 and distributed \$1.34 billion in 2024. How much Permian oil pays into it is not sourced, so the fund is context and not attribution.
who-up3	validated	institutions-commons	What two regulators did about earthquakes from wastewater disposal, from their own documents: Texas drew Seismic Response Areas, suspended 23 deep disposal permits from January 12, 2024 and cut injection around Stanton, and changed what wells must report in 2022 and 2023; New Mexico adopted its own distance-based protocol in November 2021. Whether any of it worked is not this row's question.
who-up4	validated	institutions-commons	
oil-c1	validated	energy-biophysical-economics	Texas and New Mexico's share of US crude production fell from 32 percent in 1981 to 23 percent in 2002 and then rose to 59 percent in 2025 (EIA state series); these are two states, not the Permian Basin.
seis-x1	validated	complex-systems-polycrisis	Earthquake counts fell inside Texas's seismic response areas, and fell by more in a nearby part of the same basin with no response area, so the record cannot separate the interventions from a basin-wide decline by place. Split by depth inside the treated area it partly can: deep events nearly stopped while shallow ones fell about as the basin did, held loosely on small counts and an early measurement. Whether the largest events responded cannot be tested on this record for about a century. New deep disposal capacity did not open across the state line; volumes into existing wells there are unpublished and untested.
perm-i1	integrated	author	
who-rf1	validated	institutions-commons	The federal refinery capacity report names the parent of each of Port Arthur's three refineries: Saudi Aramco (Motiva), Valero (Premcor) and TotalEnergies. Ownership only; the capacity share is another row's.
who-rf2	validated	institutions-commons	Port Arthur had 56,039 people and Jefferson County 256,526 at the 2020 Census; median household income was \$45,752 against \$76,292 for Texas, and the poverty rate 29.1% against 13.8% (ACS 2023). These describe the town that hosts the complex. They are not a count of anyone's exposure.
who-rf3	validated	institutions-commons	
who-rf4	validated	institutions-commons	
oil-c2	validated	energy-biophysical-economics	US refinery utilization climbed from 78 percent in 1985 to a peak near 96 percent in 1998 and has stayed in a narrow high band since, with idle capacity above 15 percent only in 2009 and 2020; a description of the series since 1985, not a claim about why.
oil-c3	validated	energy-biophysical-economics	Between 2019 and 2025 refinery throughput fell 1.7 percent while capacity fell 3.4 percent, so utilization is higher on less work done; the recovery after 2020 itself was demand returning, and this seat's prediction that closures drove it was tested and failed.
oil-n1	validated	energy-biophysical-economics	The seven largest US refineries are all on the Gulf Coast, which holds 54.4 percent of operable capacity, and the three in Port Arthur, Texas, together hold 7.0 percent (EIA Refinery Capacity Report 2026, calendar-day basis); why refining sits there, whether geology, ports or permits, a capacity table cannot say.
know-i1	integrated	author	
who-rf5	validated	institutions-commons	In 2021 a Port Arthur community group and two legal organizations filed a civil-rights complaint against the state environmental regulator over its oversight of a petroleum-coke facility, not against a refinery. EPA investigated for two years, the state withdrew from

			negotiations, and the file closed without a finding either way and can be re-filed. West Port Arthur, the neighborhood named, carries no figure because nobody tabulates its boundary.
who-rf6	open	institutions-commons	<i>question asked:</i> Test whether the town's largest SO2 source is the facility with the OLDEST PERMIT rather than the largest plant
study-g1	validated	institutions-commons	At four links in this study, the instrument that lets the people living there see a harm was built after the harm was established, and was built public. Four, not five: New Mexico's case is the Texas one seen from the other side of a line. 'Cannot be unpublished' holds for records that are filed and not for services that are merely kept running. At Guixi no such instrument exists at all.

salt · 32 rows

ROW	STANDING	VALIDATOR	WHAT IT HOLDS
brine-f1	validated	econ-stats-referee	The United States produces about 40 million tonnes of salt a year, brine the cheapest form at about ten dollars a tonne against fifty-six for rock salt (US Geological Survey); the roughly 14 million tonnes going to chlorine makers is derived from the survey's shares, and the survey names no individual operations.
brine-f2	validated	institutions-commons	
brine-g1	validated	institutions-commons	The Gulf belt's brine and dome operators, and the parishes and counties over the domes, named as classes because the federal survey names no operation. Wellfield and plant roles; no count. A class standing in until an operation is named.
chl-c1	validated	energy-biophysical-economics	Electricity to make a tonne of chlorine by membrane electrolysis, and the European industry's annual draw (Euro Chlor). No world share is established.
chl-f1	validated	complex-systems-polycrisis	Electrolysis makes chlorine and caustic soda in a fixed ratio, about 1.1 tonnes of caustic for each tonne of chlorine on the industry association's figures. What makes that lock bite is a storage asymmetry: chlorine cannot wait and caustic can, so production follows chlorine demand. This seat validates the mechanism; the ratio is the energy seat's reading of the primary.
chl-f2	open	complex-systems-polycrisis	<i>question asked:</i> chlorine does not ship well (liquefaction, rail tank cars, storage limits) -> chlor-alkali co-locates with its consumers; the geography row that explains the Gulf belt
chl-g1	validated	institutions-commons	Chlor-alkali producers named from a federal primary: OxyChem at Geismar; Westlake at Plaquemine, Geismar and Calvert City. And that the property-line monitoring these plants were to begin in July 2026 was deferred for two years by a proclamation filed the morning it took effect, whose Annex I lists the plants exempted.
pvc-f1	sourced	author	<i>question asked:</i> the PVC path to medical film: resin -> plasticized film (DEHP/alternatives row) -> bag; name the film/bag material grade; the copper cross-edge (wire insulation) ...
pvc-g1	validated	institutions-commons	The communities beside the vinyls stage, from the public record: an operating plant named ahead of an unbuilt one; the Formosa St. James permits characterized by venue and status (vacated 2022, reinstated January 2024, further review denied September 2024) and never as resolved; Denka excluded because it makes chloroprene, not PVC. Mossville, founded by free Black families and bought out under a 2013 voluntary purchase program that was Sasol's and not this chain's. The program's terms and valuations are not established.
nc-c1	validated	econ-stats-referee	The American Hospital Association stated that the North Cove plant made approximately 60%, or 1.5 million bags a day, of the intravenous solutions used in American hospitals; the figure is the association's alone, its origin is not given, and the company's own filing confirms the closure but not the share.
nc-f1	validated	complex-systems-polycrisis	Baxter's North Cove plant closed on September 29, 2024 on the company's own record, and the company's dated updates give its return: the first line at 32 days, all ten lines at 121, pre-storm output at 141. The quartz mine closed by the same storm never published a full-capacity date, so the two recoveries compare restarts to restarts only.

helene-f1	validated	complex-systems-polycrisis	On the National Hurricane Center's report, the core of Hurricane Helene brought catastrophic flash flooding to the western North Carolina region that holds both of this study's sites on the morning of September 27, 2024, after two days of rain that had already saturated the ground. That morning is the date of record for the flooding's arrival, at the precision of the region: the hour it reached either site is not on record.
helene-t1	validated	complex-systems-polycrisis	The National Hurricane Center's best track for Helene gives 25 positions of the storm's center, at the times the report lists, from September 23, 2024, to the evening of the 28th; a 26th dated row marks dissipation and carries no position. The line the map draws is where the center was, not where the rain fell: the center passed well to the west of both valleys, and the flooding that closed them came from the storm's rain over ground the days before had already saturated.
helene-r1	validated	complex-systems-polycrisis	NOAA's daily rainfall analysis, summed over the four days from noon on September 24 to noon on the 28th, puts 250 millimeters or more over both of this study's valleys and 500 or more over a core to their south and west, peaking at about 650 millimeters in the analysis. That is where a gauge-corrected radar analysis puts the rain, cell by cell, and not a measurement at any place: at the storm's record gauge the analysis reads about two-thirds of what the gauge caught.
nc-f2	validated	complex-systems-polycrisis	The same firm lost a single-point saline plant to a hurricane twice in seven years, Puerto Rico in 2017 and North Carolina in 2024, and both times the storm hit a supply already listed as short: the regulator's own database, archived three weeks before Helene's landfall, shows saline in shortage.
nc-g1	validated	institutions-commons	Baxter International operates North Cove, with more than 2,500 employees by two separate accounts. FDA says in its own words that it has no authority over allocation or distribution. HHS and FDA assert, with no magnitude, that group-purchasing contracting contributes to thin hospital buffers; the statutory safe harbor is what makes that contracting lawful. No primary gives hospitals' days of inventory, and none is printed.
nc-w1	validated	complex-systems-polycrisis	How long hospital stocks lasted renders as an observed interval and not as a days-of-inventory figure: it was days from the plant's closure to the first documented allocation of supplies. What patients experienced is recorded only through a survey of their pharmacists, qualitatively. The patient-impact wording is the Institutions & Commons seat's.
conv-s1	validated	energy-biophysical-economics	
route-s1	validated	energy-biophysical-economics	
route-d5	validated	energy-biophysical-economics	The finished bag's last leg is at least about 850 miles (1,350 km) in a straight line, from the plant at Marion to the Dallas area; the route itself is longer.
nc-f3	open	econ-stats-referee	<i>question asked:</i> HHS Secretary letter referenced by the AHA Oct-9 bulletin - the next primary for the 60% share
nc-g2	validated	institutions-commons	Between Hurricane Maria and Hurricane Helene, from the primary record, the response got faster: in 2024 the same FDA toolkit brought in imports under temporary authorization from nine of Baxter's own plants abroad, the first three weeks after the storm; by the end of the year, in the company's own words, nearly 200 airplanes had delivered nearly 18,000 tons of product; and one Defense Production Act order was used. What did not change: the firm's redundancy plan stayed firm-held and inspected only as appropriate, a proposal to pay hospitals for buffer stock was solicited and not adopted, the strategic reserve excludes commercial products, a six-month reserve bill was unenacted in October 2025, and the 2024 federal paper on shortages does not mention saline. Institutions learned to respond; they did not learn to be less fragile.
resp-i1	integrated	author	
bed-f1	validated	econ-stats-referee	

bed-f2	sourced	author	<i>question asked:</i> stage 6: single-use medical plastics -> regulated medical waste -> incineration (dioxin history = the PVC disposal rabbit hole, one card); the honest non-cycle: ...
bed-g1	validated	institutions-commons	The people at the bedside, split by relation: pharmacists, nurses and supply staff appear through their professional bodies' public record; patients appear only as a share in a survey of their pharmacists; the purchasing structure decides and is cited, not restated. The record holds no patient's own voice, and this study adds none.
salt-x1	validated	complex-systems-polycrisis	One storm closed a saline plant and a quartz district roughly fifteen to twenty kilometers apart in the same week. A shared hazard, not one chain reaching the other: different owners, different regulators, different customers. One observation held in two registers, not two.
salt-x2	validated	complex-systems-polycrisis	The house wire the copper chapter follows is insulated and jacketed in PVC, on the maker's own specification and the listing standard. The salt chain joins the copper chain there at a kind of material, not at a facility: nothing says any particular PVC works supplies the wire mill. Whether a chlorine disruption would stop wire is not established.
salt-x3	validated	complex-systems-polycrisis	
rh-chl-rail	open	institutions-commons	<i>question asked:</i> chlorine on rails: the regulation-history card (public record only, per guardrail)
rh-wfi-water	open	energy-biophysical-economics	<i>question asked:</i> the OTHER ingredient: water-for-injection — the bag is salt + water + power; one card on WFI stills (energy comparison per standard)
rh-maria-2017	validated	institutions-commons	

sand · 34 rows

ROW	STANDING	VALIDATOR	WHAT IT HOLDS
spine-c1	validated	econ-stats-referee	No agency publishes Spruce Pine's share of the world's high-purity quartz, and the 70 to 90 percent figures in circulation trace to industry analysts; what is validated is that absence and the way the chapter renders it, with the counter-view in its own voice, and no share number is established.
spine-f1	validated	complex-systems-polycrisis	Sibelco, one of Spruce Pine's two quartz operators, halted on September 26, 2024 and announced a restart of production and shipments about fourteen days later, on its own statements, with ramp-up continuing. Neither operator published a date for full capacity, so comparisons with this clock render restarts and first lines only.
spine-f2	validated	energy-biophysical-economics	High-purity quartz forms the inner layer of the crucible that holds molten silicon while a crystal is pulled, not the whole vessel, and crucibles for semiconductor and for solar crystal-pulling are different markets.
spine-g1	validated	institutions-commons	Spruce Pine's two operators: Sibelco, and The Quartz Corp, a 50/50 Imerys and Norsk Mineral venture with its purification plant in Norway. The town had 2,194 people and the county 14,903 at the 2020 Census. A 2025 union representation petition at Sibelco's Crystal Plant ended with no union certified, the tally unverified. Quartz carries no federal critical-mineral designation.
spine-w1	validated	complex-systems-polycrisis	Both Spruce Pine operators said in their own words that feedstock and finished stocks held elsewhere in the chain would carry their customers through the closure, and one called that slack a deliberate choice made after the pandemic. No quantity of cover is stated. The buffers bought the weeks; whether substitution would have bought months was never tested.
spine-c2	validated	econ-stats-referee	
spine-i1	integrated	author	

route-d2	validated	energy-biophysical-economics	The sea distance between ports of the kind each end of the sand chain's Pacific leg would use: 4,839 to 5,185 nautical miles (8,962 to 9,603 km, or about 5,600 to 6,000 miles), Yokohama or Kobe to Los Angeles (NGA Pub. 151); anchored at both ends, since the study names neither the wafer works nor the US port, and not a measurement of this cargo's voyage.
route-t4	sourced	energy-biophysical-economics	<i>question asked:</i> Days at sea for the sand chain's Pacific leg (wafers, a Japanese port to a US west-coast port).
route-f1	validated	energy-biophysical-economics	
cru-f1	validated	author	A named fused-quartz crucible maker, at company level: Momentive Technologies, four Ohio plants, by its own account. No plant is identified as making the crucible this chain follows.
cru-c1	open	econ-stats-referee	<i>question asked:</i> crucible consumption: CZ crucibles are single-use-ish (one or few ingot pulls per crucible) -> quartz demand scales with wafer output; needs an industry primary
cru-g1	validated	institutions-commons	Who: the makers of fused-quartz crucibles are specialist firms. One is named in this study, at company level only; no plant is identified as making the crucible this chain follows. The work is furnace operation, forming and quality control. No headcount is given, and this study holds nothing on the communities beside these works. A class, standing in until a plant is named.
poly-f1	validated	energy-biophysical-economics	Solar-grade polysilicon takes about 49 kWh of electricity per kilogram (IEA PVPS), which puts the silicon in one small chip at about a watt-hour; the electronic-grade figure is higher and unpublished, and the chip fab's own electricity, the larger term, is a figure this study named and did not pull.
poly-g1	validated	institutions-commons	Who: polysilicon producers are chemical manufacturers and heavy users of electricity, as the sand chapter's own figure for them shows. This study names none of them and has not sourced where their plants are. The work is process operation, maintenance and electrical trades. No headcount is given, and this study holds nothing on the communities beside these plants. A class, standing in until a plant is named.
waf-f1	validated	econ-stats-referee	SUMCO calls itself a front-runner in silicon wafers on its own website, and its one published share, more than half, is of wafers for leading-edge logic chips only; we found no public agency that publishes wafer makers' shares, and no share of the wafer market is established.
waf-g1	validated	institutions-commons	Who: the firms that pull silicon crystals and slice them into wafers. One, Japan's SUMCO, describes itself as among the front-runners; no public agency publishes the shares, and this study has not sourced where the works are. The work is crystal pulling, slicing and measurement. No headcount is given, and this study holds nothing on the communities beside these works. A class, standing in until a plant is named.
sand-s1	validated	author	
fab-f1	validated	complex-systems-polycrisis	After the chip shortage a US maker announced and began a large analog-chip plant in Texas, with about three years between announcement and first output: a capacity response in the same shape the transformer queue model describes. The wider claim, that older-generation chip capacity went unbuilt before 2020, is only partly sourced, and the chapter's own aside says so.
fab-f2	validated	econ-stats-referee	The Commerce Department's 2021 inquiry, drawing more than 150 responses, found the median inventory of the chips buyers highlighted had fallen from 40 days in 2019 to under 5 in 2021, and named analog chips used in power management among them; it is a median across respondents, not a measure of any one chip.
fab-f3	open	econ-stats-referee	<i>question asked:</i> PMIC lead-time/price record 2021-22 (40-52+ wk lead times; +10% 1H22 on 2021 rises) — trade-record primaries (distributor lead-time data if obtainable); the Apple ...
fab-f4	validated	econ-stats-referee	

fab-g1	validated	institutions-commons	Texas Instruments signed an award agreement in December 2024 for up to \$1.6 billion in federal CHIPS funding across three fabs in two states; an award agreement is not a disbursement. The company's own forward statements: as many as 3,000 direct jobs at a four-fab Sherman build, and 2,000 company jobs across the three funded fabs. Promises in the company's voice, never combined. Local incentives and the community beside the site are not on this study's record.
fab-g2	open	institutions-commons	<i>question asked:</i> Grayson County / Texas state-local incentives (Chapter 313 / JETI agreements, county abatements) and the community record around the Sherman fabs. Split from fab-g1.
fab-i1	integrated	author	
dev-f1	validated	econ-stats-referee	
ew-f1	validated	complex-systems-polycrisis	The federal commodity summaries report that post-consumer scrap provided about 150,000 tonnes of US copper in 2024, against 720,000 tonnes from manufacturing scrap, and report silicon recycling in one word: insignificant. So the copper on a circuit board can return and the silicon does not, and board copper is bounded above by that post-consumer stream and sits well below it. A ceiling, not a share.
ew-g1	validated	institutions-commons	A recovery workforce, formal and informal, exists where the world's electronics are taken apart. A class only: no count, no place, and no facility named.
sand-x1	validated	complex-systems-polycrisis	One storm closed a quartz district and a saline plant roughly fifteen to twenty kilometers apart in the same week. A shared hazard, not one chain reaching the other. One observation held in two registers, not two.
sand-x2	validated	complex-systems-polycrisis	
sand-x3	open	complex-systems-polycrisis	<i>question asked:</i> EV/lithium pull on GOES (goes-f2, iron register) is the same same-machines competition pattern: chains competing for one industrial asset — motif row
sand-x4	validated	complex-systems-polycrisis	Copper recovered from circuit boards re-enters the copper chain at its scrap pool. A join at a kind of material, never at a named smelter, and small: it renders as where the sand chain's copper goes, not as a route a disruption travels.
rh-spine-helene	open	institutions-commons	<i>question asked:</i> the storm itself: Helene's inland flood signature (NOAA/USGS record) — one card, public record, feeds sand-x1
rh-cz-energy	validated	energy-biophysical-economics	

Appendix B · engines, seeds, reproductions

Each engine is seeded and has its results and a raw-draw sample beside it. Files are named here by FILE NAME, never by a folder in our working repository, which no reader has; the table at the foot of this appendix gives each file’s fingerprint, so anyone holding the study’s published materials can confirm they hold the same file. Below, a **reimplementation** is a second hand’s own code with a different seed and nothing imported from the engine. A **probe** runs the engine’s own code and shares it. The two are not the same kind of evidence and are listed apart. Verdicts are read from the files that hold them.

E3, the transformer queue

Engine: `e3_transformer_queue_mc.py`, seed 33. Three objects were reproduced and they differ in standing:

OBJECT	STANDING	REPRODUCED BY	RESULT
the divergence boundary	current	reimplementation: <code>e3_reimpl5_seed7.py</code> , own construction, a deterministic skeleton with no draws	reproduces
the mechanical ceiling (the one-class intake probe)	current	reimplementation: <code>e3_ceiling_expectation_author.py</code> , arrivals at expectation	REPRODUCES; worst difference 0.01 month across eight rows. A check on implementation, not on the single-queue structure the ceiling is conditional on.
the same ceiling, as a Monte Carlo	current	referee cross-seed run of the Author’s engine, seed 11, in the referee’s verification package; Author, <code>e3_ceiling_mc_author.py</code> , seed 7	medians agree to 0.05 month
the σ map	superseded grid	reimplementation: <code>e3_reimpl4_seed7.py</code> , seed 7	reproduces that grid; the current map’s top member has not been reproduced outside the engine
engine behavior generally		probes (share the engine’s code): the referee’s verification scripts	referee probes of the engine, not a rebuild

E3-C, the reserve/pool companion

Engine: `reserve_pool_mc.py`, seed 41. Reimplementation: `reserve_reimpl_seed7.py`: REPRODUCES (6 legs, all pass).

In the seat’s words: INDEPENDENT compatibility OVERSTATES pooling. Real compatibility is clustered by design family. Every pooling figure here is an UPPER BOUND on the voluntary substitute’s reach. That bound is on the pool’s reach: whether some other member’s spare fits. What pooling adds is reach multiplied by the hole a holder’s own spares leave, and the same clustering moves the hole, in a direction this engine does not model. Read every gain here for its direction and its order, never for its size.

The σ fit

Engine: `eb_d039_sigma_fit.py`, with its results file beside it. **It has no seed and draws nothing:** the fit is deterministic, and re-running it returns the same file. Its own numbers render — the arrival envelope a chapter gives as “between about one in five and one in twenty” is this file’s.

This is the one engine in this study that a second hand has not reproduced. Every other engine above has been reimplemented or probed by somebody who did not write it; this one has only been re-run by its author, who reports it byte-identical. A deterministic fit re-run by its author confirms that the file on disk is what the script produces. It does not test the choices inside the script, which is what a second hand tests, and the reader should weigh it as the wide envelope it reports rather than as a settled number.

Running it

The small inputs travel with the engine: `ferc1_acct353_additions.json`, `ppi_PCU3353113353111.json` are committed and ship beside it. The three EIA-860 annual zips and the LBNL interconnection-queue workbook do not — together about 74 MB, and this study pins large raws rather than committing them. **Fetch the three EIA-860 zips and the LBNL workbook listed below into the same folder as the script, unopened, and run it; it reads the zips directly and names anything missing.**

FILE	MD5	SOURCE
eia860_2022.zip	2fbba665a021a7b4943349d72405cddd	https://www.eia.gov/electricity/data/eia860/archive/xls/eia8602022.zip
eia860_2023.zip	12ad0f63fabf0025ebf53f6728f13c18	https://www.eia.gov/electricity/data/eia860/archive/xls/eia8602023.zip
eia860_2024.zip	684352896d6a62b5d2d98a403b30e120	https://www.eia.gov/electricity/data/eia860/xls/eia8602024.zip
lbnl_ix_queue_data_file_thru2024_v2.xlsx	f67f732679a0f3d223b34e24fc84ee44	https://eta-publications.lbl.gov/sites/default/files/2025-08/lbnl_ix_queue_data_file_thru2024_v2.xlsx

Fingerprints

SHA-256 of each file named above, taken with line endings normalized to LF, so that the value is the same on every platform. Computed when this paper was built.

FILE	SHA-256
e3_transformer_queue_mc.py	ef84b1a408fddb8d9f73d9cb030637fbadbc845d54a5fd185fdc8e1b95a823e8
e3_reimpl5_seed7.py	6099fefbd340c4a949eb9a29ad0fb44c4a5633909cfaf40ac79867f112ea5db7
e3_ceiling_expectation_author.py	5dd4641956191d0b368fb838fedab2e3363e72564c91e75423408ce446a89523
e3_ceiling_mc_author.py	4548bcee417de1967cbef9faec1a178e099236596249dd9d7d965ca71428090d
e3_reimpl4_seed7.py	4c2c10500f76443d6332d48d56041c6ce19806161d3d6d71188744059c4d5922
reserve_pool_mc.py	5be1e2690146c809bb7478c9ea5ec8c7d0296dd31ca78d6c90e0aae031c8b00e
reserve_reimpl_seed7.py	d6a72fedafcf1f74e6ba529a1253f51e9485bd002339eff48b889b8f0afe0419
eb_d039_sigma_fit.py	2f37a3e0254291f1fb8edb5058c21366e3ec5ad19f623edc78e2fbdc380ef85d
eb_d039_sigma_fit_results.json	01ed4d45ff2cea420b08e26293b69bb9ae1811029b561059f6f380c8f501ee2c